

The **Curious Incident** of the **Log** in the **Downtime**

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Hello Zabbix Summit! How's everybody doing out there?

It's great to be back. I was in Riga with you this time last year, it was a lot of fun, and I was delighted when the team from Zabbix got in touch and asked if I'd write a talk specifically for this year's Zabbix Summit.

For the folks who weren't here last year, this is me.

DYLAN BEATTIE
DEVELOPER
CONSULTANT
ARCHITECT
SPEAKER
GUITAR PLAYER
STORYTELLER
...ZABBIX EXPERT?
LOL! NO.



I'm Dylan, and I do a lot of things. I'm a developer, consultant, architect, keynote speaker, guitar player, storyteller... but one thing I'm definitely not is a Zabbix expert. I suspect every single person in this room knows the details of the Zabbix platform better than I do, so if you're here to learn how to achieve 1 million NVPS or how to replace a hundred systems with one Zabbix - well, the good news is, there's some awesome talks coming up that'll tell you how to do that. No, what I like to do is to dig into the relationships between the digital world and the physical world. I like to look for patterns and parallels, stories that can help us understand one system by comparing it with another. I've been working in tech since the 1990s - one of my first jobs was "webmaster", back in the days when that meant you knew HTML, CSS, JavaScript, Linux, Apache, SQL, PHP, TCP/IP networking, how to configure a firewall... So although I've never specifically worked as a monitoring engineer, I've absolutely been the guy who gets yelled at whenever something stops working.

And so, to me, the value in monitoring and observability platforms is that they help to answer two very important questions:

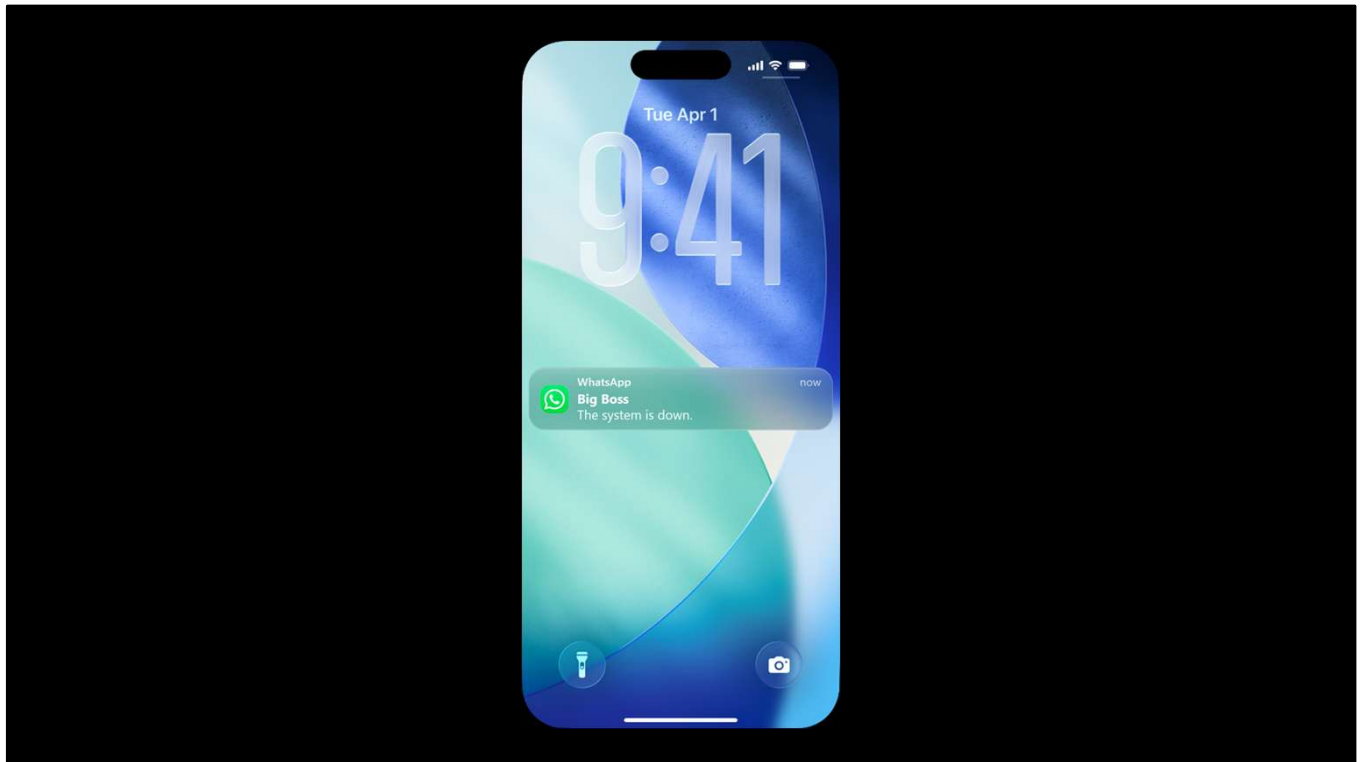
WHY MONITORING MATTERS

1. **What the **** just happened?!**
2. **...is it going to happen again?**

- What the heck just happened?
- Is it going to happen again?

OK, it's just possible that somewhere out there there's a system so well-engineered and so well-instrumented that it's never had a single production incident, in which case the question "what happened" isn't really relevant... but if such a system does exist, it's definitely not one of the ones I've ever worked with. I've worked on a handful of systems over my career where we'd maybe come close to achieving that level of intelligence, and most of the time we could anticipate bad things before they actually happened, and that way we could prevent them from happening.

Dylan Beattie: The Curious Incident of the Log in the Downtime.



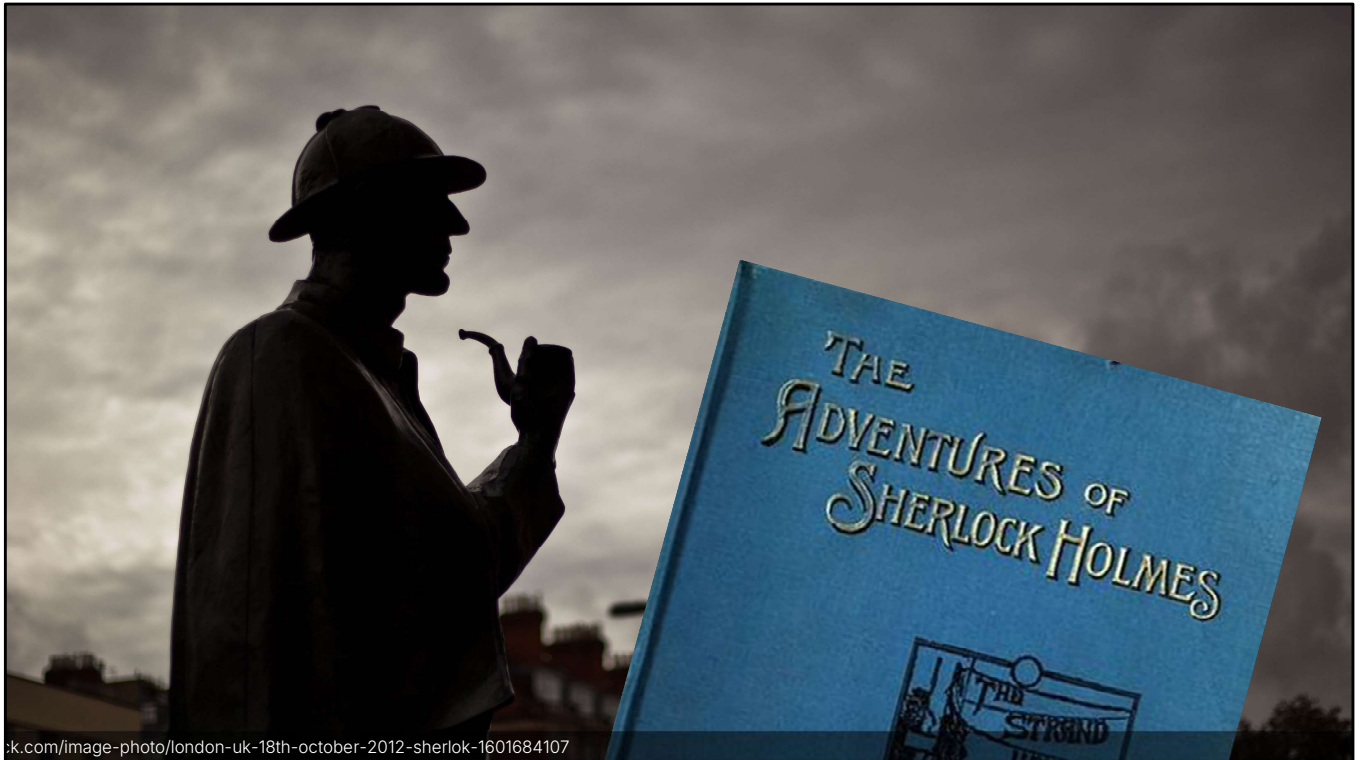
But I've got to be honest: for every incident that we avoided, there were probably a dozen where we had no idea anything was wrong until that first notification popped up - "the system is down".

I love that phrase. "The system is down". It's so helpful. Which system? You know, THE system. What do you mean "down". I mean it's not there! Because when all the ticket says is "the system is down"... well, now you're not an engineer any more. No, turns out that today, you're going to be a detective.

Dylan Beattie: The Curious Incident of the Log in the Downtime.



You've got to interview the witnesses: which system were you using? What were you trying to do? Does it work if you try it on your phone? You've got to go hunting for clues - pinging IP addresses, reading through log files, digging through stack traces... and eventually, if you're lucky, the clues will lead you to the answer. And as I was thinking about this, it occurred to me: people love detective stories. Books, films, TV - we just love a good mystery, and we love to watch as our detective hero deploys their powers of deduction to solve the mystery, unmask the murderer, or figure out what happened to the stolen secret plans. But what if solving crimes isn't just about observation... what if it's about observability? To help us figure this out, let's bring in a consultant.

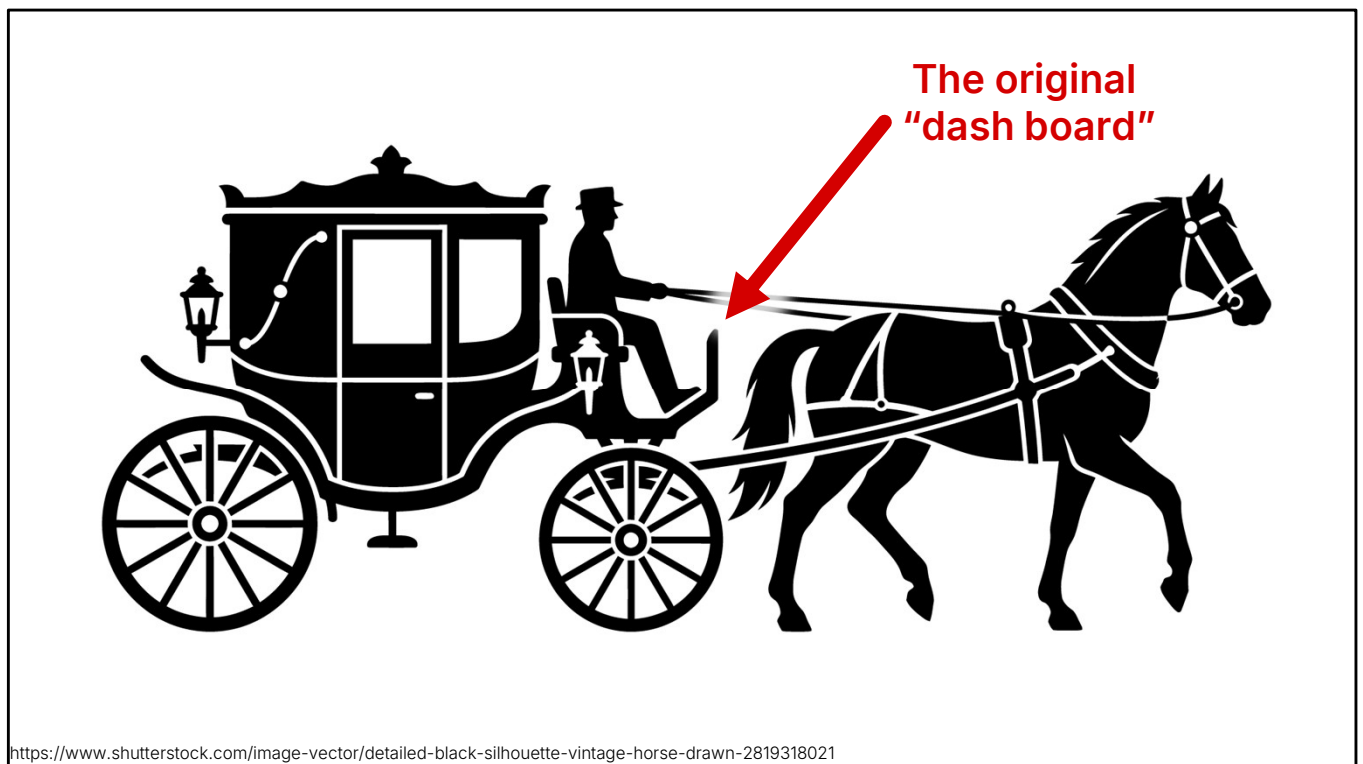


The greatest consultant of them all.... Sherlock Holmes. Created by Sir Arthur Conan Doyle, the Sherlock Holmes stories have been adapted countless times across films, television, video games - and yes, Sherlock is a consulting detective. He's the expert you bring in when nobody can figure out what's going on.

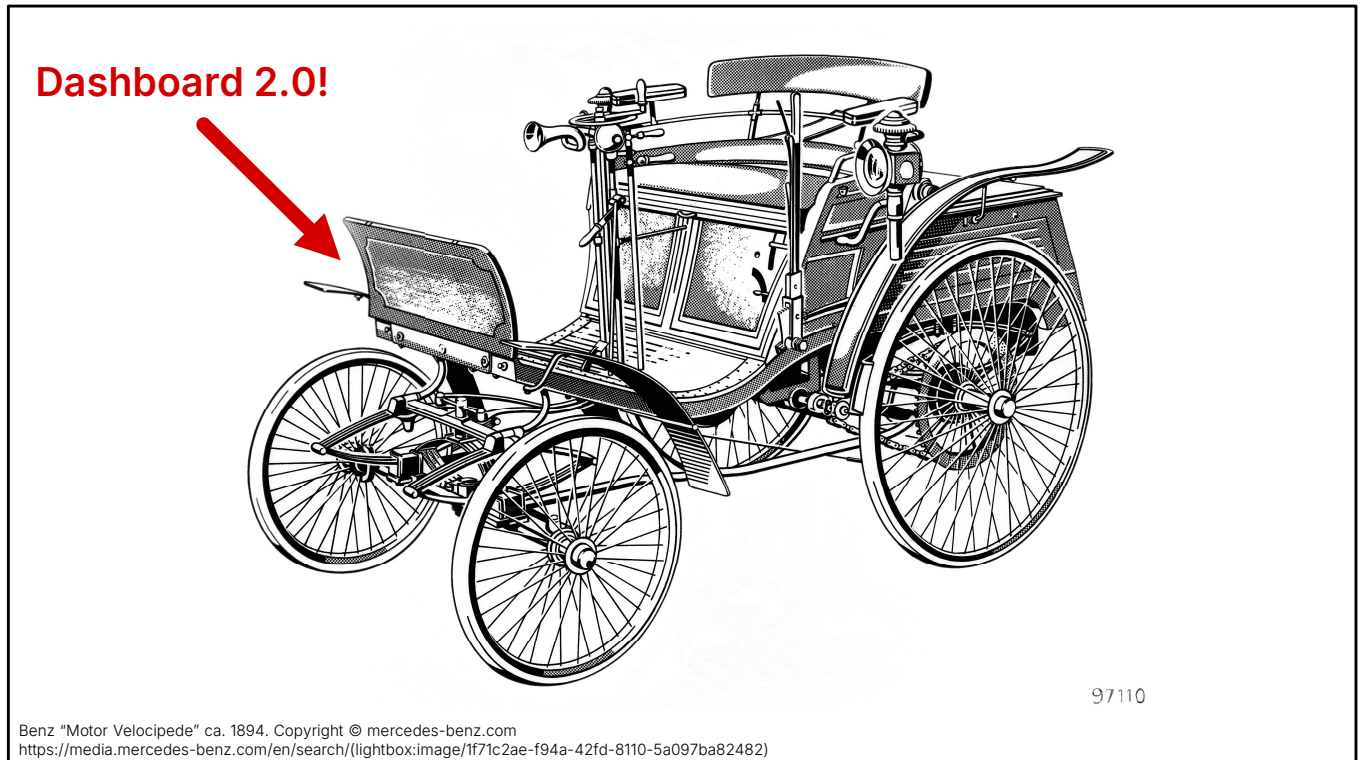


Old Victorian London" by Arnaud Imobersteg / <https://www.artstation.com/artwork/8xlvO>

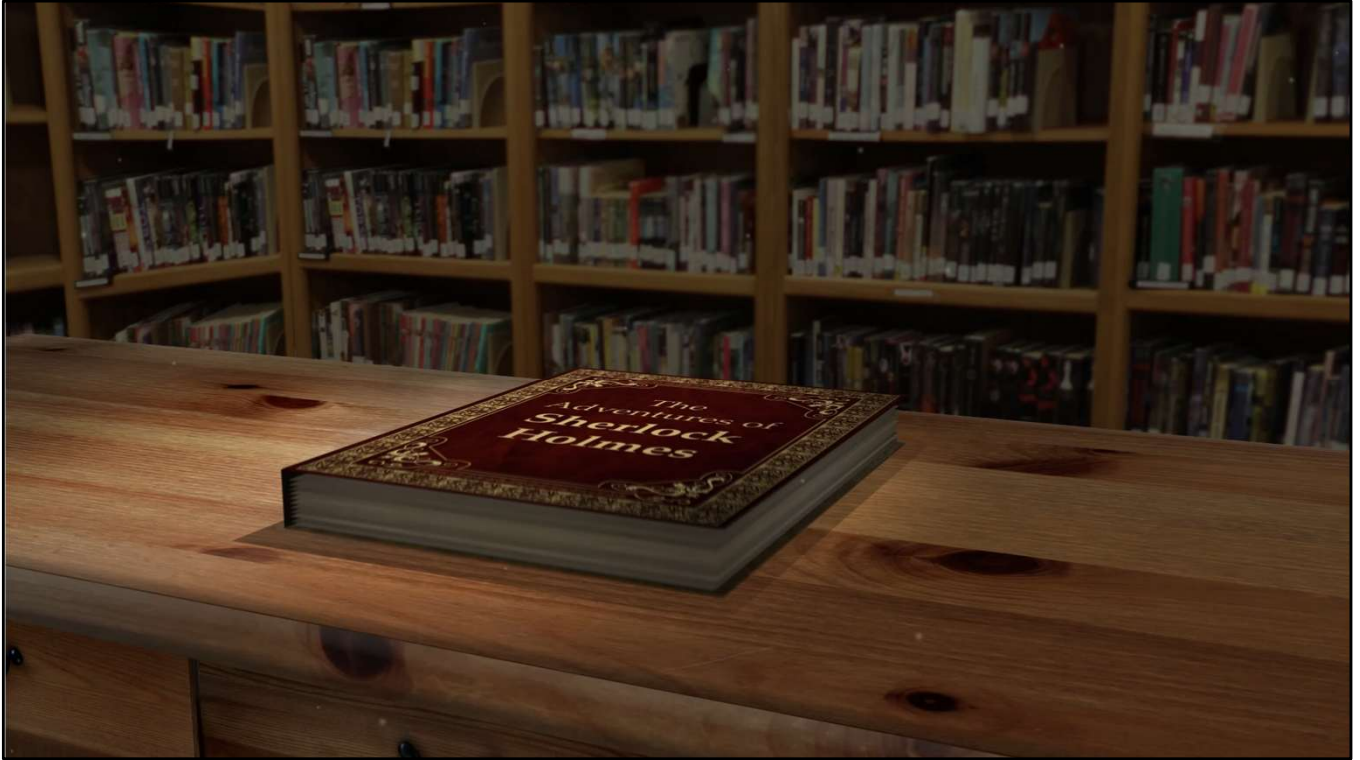
But to a modern reader, the setting itself, Victorian London, is as much the star of the stories as Sherlock's brilliant deductions. Now, we're talking here about the last years of the nineteenth century, around 1880 to 1900. There's a telegraph network, we've had film cameras for a few decades, the telephone is still a pretty neat idea - and although motor cars and computer networks are still many decades away, there is one beautifully anachronistic word that was already in widespread use



Victorian London ran on horse-drawn carriages - if you were rich enough to afford one. The passengers rode in the relative comfort of the carriage while the driver sat up front, and the horses' hooves would throw up, or dash, stones, mud, and whatever else was lying around the streets of London - and so a wooden board was installed in front of the driver. This was the original dashboard.



When motor cars were first invented a few decades later, the wooden panel still protected the driver from mud and stones, but also provided a convenient place to attach controls and instruments - and then, as we started to add telemetry and instrumentation to software systems, we borrowed the same word. But it's maybe nice to remember that for nearly two hundred years, a good dashboard has been a thing that let you move fast without getting covered in horse manure.



But when it comes to observability and telemetry? Well, Victorian London provided some challenges...

such as in the Adventure of the Six Napoleons.

Sherlock Holmes is called to investigate a series of bizarre burglaries: somebody has been breaking into houses all over London, not to steal anything, but to destroy busts of Napoleon.

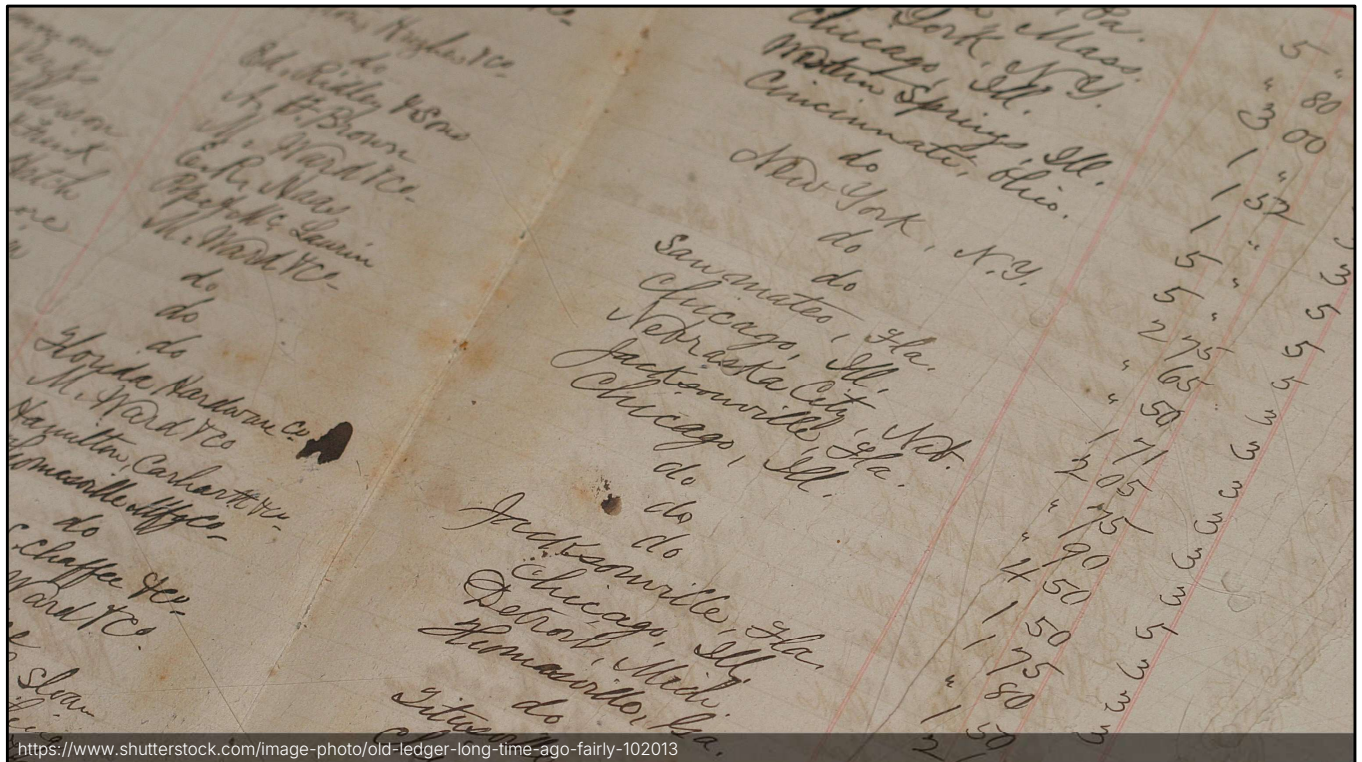


And not just any old busts of Napoleon - the busts that are destroyed are all apparently identical. Given the vast number of statues of Napoleon displayed around London, Sherlock concludes that it's too much of a coincidence that anybody who just set out to smash up statues because they hated Napoleon would destroy three identical specimens... and so he turns to the log files.

wait, log files? They had log files in Victorian London?

Sure they did. Not electronic files - those won't be invented for a while - but they had files.

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Hand-written paper ledgers, ordered by date. Thanks to those canny Victorians and their excellent logging infrastructure, Sherlock is able to find out where the statues were purchased, and determine that they all came from a single batch of six Napoleons, produced in the same factory on the same day... by the same man, an Italian sculptor called Beppo.

THE ADVENTURE OF THE SIX NAPOLEONS

nd once, when my hand seemed within his reach, he snapped at it like a hungry wolf. We stayed long enough at the police-station to learn that a search of his clothing revealed nothing save a few shillings and a long sheath knife, the handle of which bore copious traces of recent blood. "That's all right," said Lestrade, as we parted. "Hill knows all these gentry, and he will give a name to him. You'll find that my theory of the Mafia will work out all right. But I'm sure I am exceedingly obliged to you, Mr. Holmes, for the workmanlike way in which you laid hands upon him. I don't quite understand it all yet."

"I fear it is rather too late an hour for explanations," said Holmes. "Besides, there are one or two details which are not finished off, and it is one of those cases which are worth working out to the very end. If you will come round once more to my rooms at six o'clock to-morrow I think I shall be able to show you that even now you have not grasped the entire meaning of this business, which presents some features which make it absolutely original in the history of crime. If ever I permit you to chronicle any more of my little problems, Watson, I foresee that you will enliven your pages by an account of the singular adventure of the Napolconic which he was wont to assume. At last he started in

Holmes listened with polite attention; but I, who knew him so well, could clearly see that his thoughts were elsewhere, and I detected a mixture of mingled uneasiness and expectation beneath that mask



"HIS NAME, IT APPEARED, WAS BEPPO"

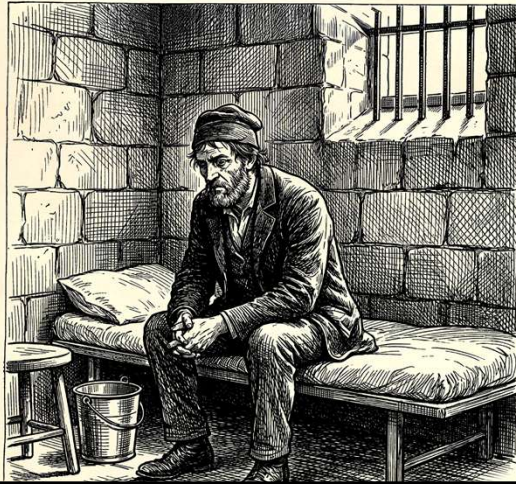
And, thanks to the factory's employee records - more log files - ascertain that Beppo vanished right after the statues were made... he'd come running into the factory one day, pursued by the police, who had arrested him, and Beppo had spent the last year in jail.

THE ADVENTURE OF THE SIX NAPOLEONS

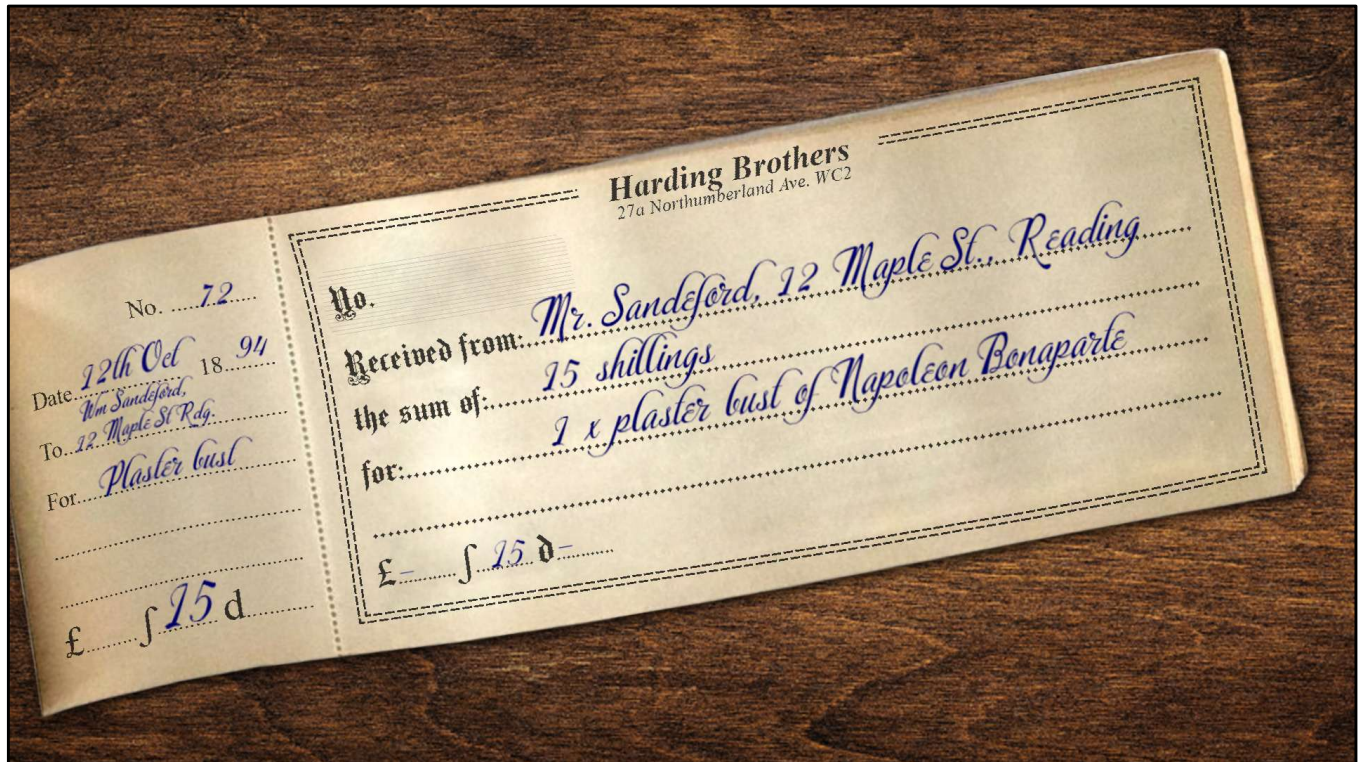
"Yes, gentlemen," said he, "it is the most famous pearl now existing in the world, and it has been my good fortune, by a connected chain of inductive reasoning, to trace it from the Prince of Colonna's bedroom at the Dacre Hotel, where it was lost, to the interior of this, the last of the six busts of Napoleon which were manufactured by Gelder & Co., of Stepney. You will remember, Lestrade, the sensation caused by the disappearance of this valuable jewel,

the sequence of events, though you see them, of course, in the inverse order to the way in which they presented themselves to me. Beppo had the pearl in his possession. He may have stolen it from Pietro, he may have been Pietro's confederate, he may have been the go-between of Pietro and his sister. It is of no consequence to us which is the correct solution. "The main fact is that he had the pearl, and at that moment, when it was on his person, he was pursued

by the police. He made for the factory in which he worked, and he knew that he had only a few minutes in which to conceal this enormously valuable prize, which would otherwise be found on him when he was searched. Six plaster casts of Napoleon were drying in the passage. One of them was still soft. In an instant Beppo, a skilful workman, made a small hole in the wet plaster, dropped in the pearl, and with a few touches covered over the aperture once more. It was an admirable hiding-place. No one could possibly find it. But Beppo was condemned to a year's imprisonment, and in the meanwhile his six busts were scattered over London. He could not tell which contained his treasure. Only by breaking them could he see. Even shaking would tell him nothing, for as the plaster was wet it was probable



Sherlock notices that the incident happened two days after a fabulously rare black pearl disappeared from a London hotel, and deduces that Beppo must have stolen the pearl, and while running away from the police, he'd run into the factory and hidden the pearl inside one of the plaster busts of Napoleon immediately before he was arrested.

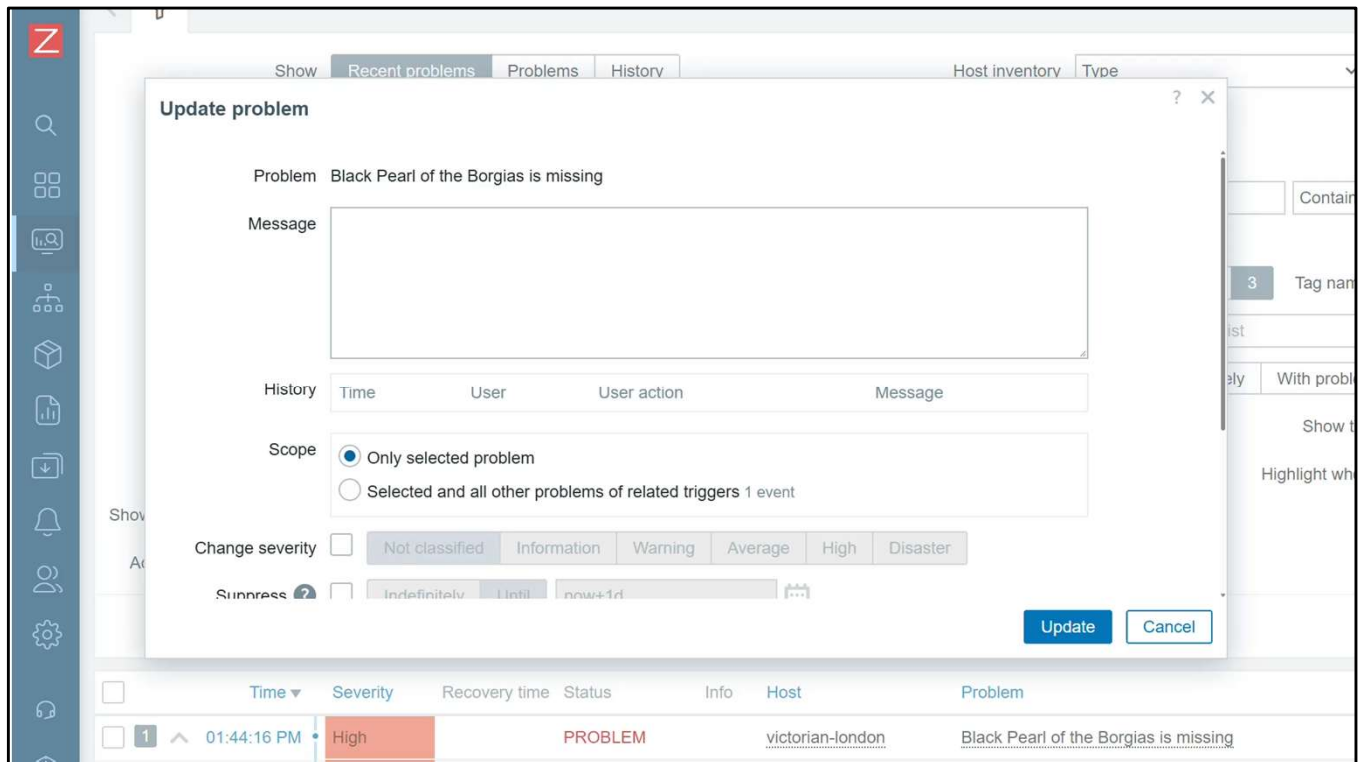


And then, in the piece de resistance, Sherlock checks more log files - from the factory's sales ledger, he finds out which shop bought the rest of that batch; from the store's receipt book, he finds out who owns the final remaining statue - which is, of course, a gross violation of the GDPR, but that also hadn't been invented yet - and then writes to the owner saying he's very keen to acquire his statue of Napoleon and will pay him ten pounds for it - a small fortune at that time. They make a deal, the oblivious owner hands over his statue and walks away ten pounds richer, and the story ends with a victorious Sherlock Holmes smashing the statue open to reveal the priceless



Black Pearl of the Borgias, and so we can close the incident:

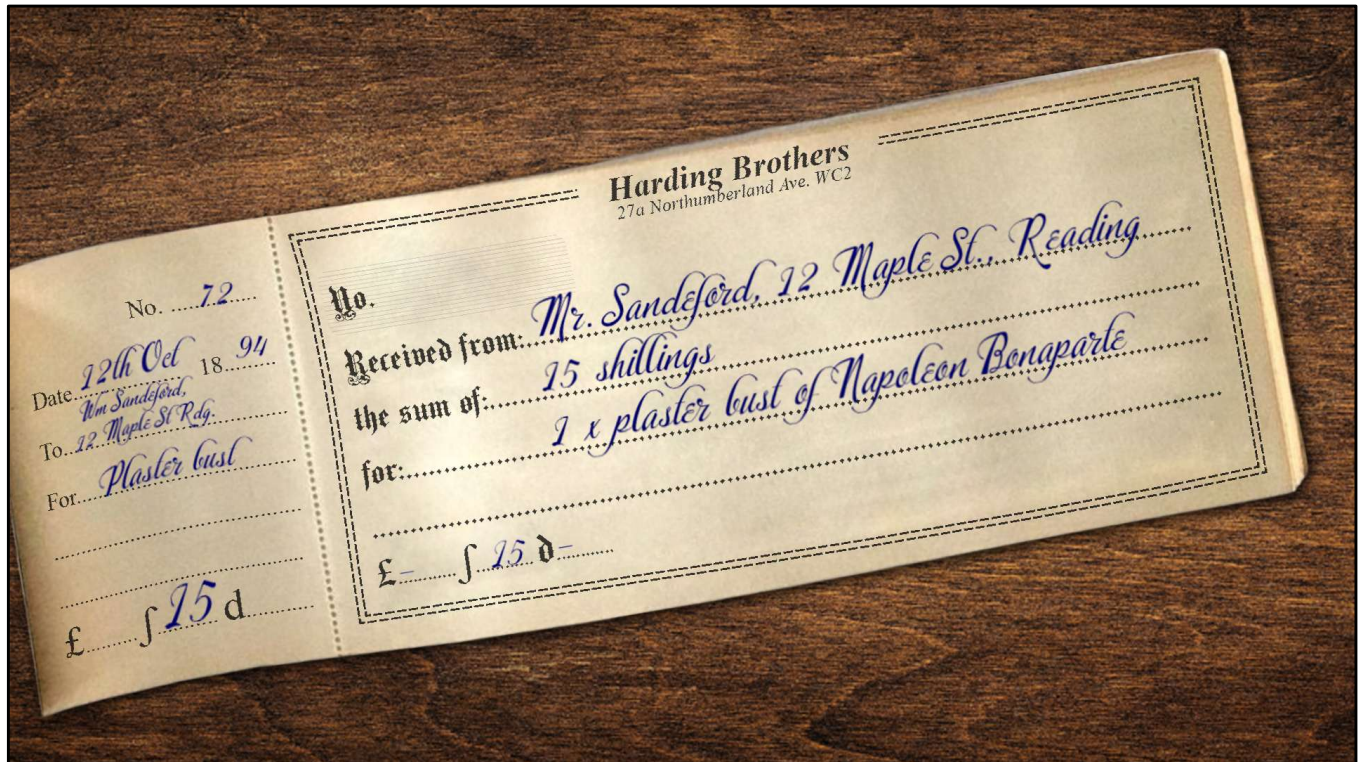
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Found by sherlock holmes inside a plaster bust of Napoleon

Resolved - and it's all thanks to log files.

Well... kinda. The thing about log files is they only record the things that somebody decided to record. In the case of the Six Napoleons, the details which cracked the case were records of business activities:



a shop recording their sales, a factory keeping employee records. There have been societies in human history which kept far more extensive logs - usually because they were under the control of totalitarian regimes trying to track and monitor every movement of their citizens, even to the extent of installing listening devices in private homes and offices; we'll talk more about that a little later - but in Victorian London, by and large, citizens were free to go about their lives unrecorded and undocumented, unless they were involved in some kind of business transaction. And, of course, if any of those sales ledgers or employee records had been lost, or destroyed, Sherlock may never have cracked the case.

I've spent a good chunk of my software development career reading log files. When I first started building applications, back in the late 1990s, the log files were often the only thing you had to go on when trying to figure out what caused a production incident.

The Thing about Logs...

1. Where do you store them?
2. How long to you keep them?
3. How do you keep them safe?
4. ...do they even contain the right stuff?

Log files present four distinct challenges to developers and support engineers. First: you've got to figure out where to keep them. A log file is no use if it's on the hard drive of a remote server that's not responding to pings any more. Second: you've got to figure out how long to retain them for, otherwise they're just going to keep growing until you run out of disk space. Third: You've got to secure them: log files could contain all sorts of information that could help an attacker or an adversary gain access to one of your systems.

But the biggest challenge with log files is that you can only see the things that the developers decided to log - and that might be a decision made years ago, by somebody you've never met, who had absolutely no idea where their software would end up running or how it was going to get used.

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```
2026-09-21 09:14:02,117 INFO web.LoginController - Login attempt for user 'jsmith' from 10.4.12.88
2026-09-21 09:14:02,241 FATAL auth.PasswordValidator - Password mismatch for user 'jsmith'. CANNOT CONTINUE.
2026-09-21 09:14:02,242 FATAL auth.PasswordValidator - Hash mismatch (user typed 'Summer2026')
2026-09-21 09:14:02,243 INFO web.LoginController - Redirecting 'jsmith' to /login?error=1
2026-09-21 09:14:05,880 INFO web.LoginController - Login attempt for user 'jsmith' from 10.4.12.88
2026-09-21 09:14:05,991 INFO auth.PasswordValidator - Password OK for user 'jsmith'
2026-09-21 09:14:05,994 WARN data.ConnectionPool - Could not obtain connection from pool 'primary'
2026-09-21 09:14:05,995 WARN data.ConnectionPool - Failing over to pool 'secondary'
2026-09-21 09:14:06,002 WARN data.ConnectionPool - Could not obtain connection from pool 'secondary'
2026-09-21 09:14:06,003 WARN data.SessionRepository - Session store unavailable; continuing with session=null
2026-09-21 09:14:06,010 INFO web.DashboardController - Rendering dashboard for user 'null'
2026-09-21 09:14:06,044 WARN data.OrderRepository - Database unavailable; returning empty order list
2026-09-21 09:14:06,045 WARN data.InvoiceRepository - Database unavailable; returning empty invoice list
2026-09-21 09:14:06,046 WARN data.BalanceRepository - Database unavailable; assuming account balance = £0.00
2026-09-21 09:14:11,302 INFO web.LoginController - Login attempt for user 'mpatel' from 10.4.19.201
2026-09-21 09:14:11,398 FATAL auth.PasswordValidator - Password mismatch for user 'mpatel'. CANNOT CONTINUE.
2026-09-21 09:14:11,399 FATAL auth.PasswordValidator - Caps Lock may be on. System integrity compromised.
2026-09-21 09:14:12,650 WARN jobs.NightlyBillingJob - Database unavailable; skipping billing run for 14,302 customers
2026-09-21 09:14:12,651 INFO jobs.NightlyBillingJob - Billing run completed in 1ms
2026-09-21 09:14:15,007 WARN health.HealthEndpoint - db-prod-01: DOWN, db-prod-02: DOWN
2026-09-21 09:14:15,008 INFO health.HealthEndpoint - Overall status: HEALTHY (0 FATAL events in last 60s excluding auth)
```

Worse - most logging libraries offer you a choice of logging levels. That developer had to decide - hey, is this a Fatal error? Or an Error, or a Warning? You ever worked on a system where the developers decided that the user entering the incorrect password was Fatal, but the database being offline was just a Warning?

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```
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2026-09-21 09:14:05,994 WARN data.ConnectionPool - Could not obtain connection from pool 'primary'
2026-09-21 09:14:05,995 WARN data.ConnectionPool - Failing over to pool 'secondary'
2026-09-21 09:14:06,002 WARN data.ConnectionPool - Could not obtain connection from pool 'secondary'
2026-09-21 09:14:06,003 WARN data.SessionRepository - Session store unavailable; continuing with session=null
2026-09-21 09:14:06,010 INFO web.DashboardController - Rendering dashboard for user 'null'
```

Worse - most logging libraries offer you a choice of logging levels. That developer had to decide - hey, is this a Fatal error? Or an Error, or a Warning? You ever worked on a system where the developers decided that the user entering the incorrect password was Fatal, but the database being offline was just a Warning?

```
log.Fatal Wake_Up_My_Boss_At_4AM_On_A_Sunday()
```

```
Log.Error Apologize_And_Raise_A_Ticket()
```

```
log.Warn Alert_If_This_Happens_1000_Times()
```

```
log.Info Everything_Is_Fine_Just_Checking_In()
```

```
log.Debug Hide_The_Bug_In_100Gb_Of_Stack_Traces()
```

Personally, I think we should have given them better names: Instead of `log.Fatal`, it should be `log.WakeUpMyBossAt4amOnSunday`. Instead of `log.Error`, it should be `log.SaySorryAndOpenATicket`. `log.Warn` should be `log.TellMeIfThisHappensOneThousandTimesInARow`, `log.Info` should be `log.EverythingIsFineJustCheckingIn`, and `log.Debug` should be `log.FillUpMyDiskWithStackTraces`.

So, that's logging: logging is all about keeping records, and it's only effective when the person who built the system made the right decisions about what to log, how much detail to capture, where to store those logs, and so on...

```
log.Fatal Wake_Up_My_Boss_At_4AM_On_A_Sunday()  
  
Log.Error Apologize_And_Raise_A_Ticket()  
  
log.Warn Alert_If_This_Happens_1000_Times()  
  
log.Info Everything_Is_Fine_Just_Checking_In()  
  
log.Debug Give_Me_100Gb_Of_Stack_Traces()
```

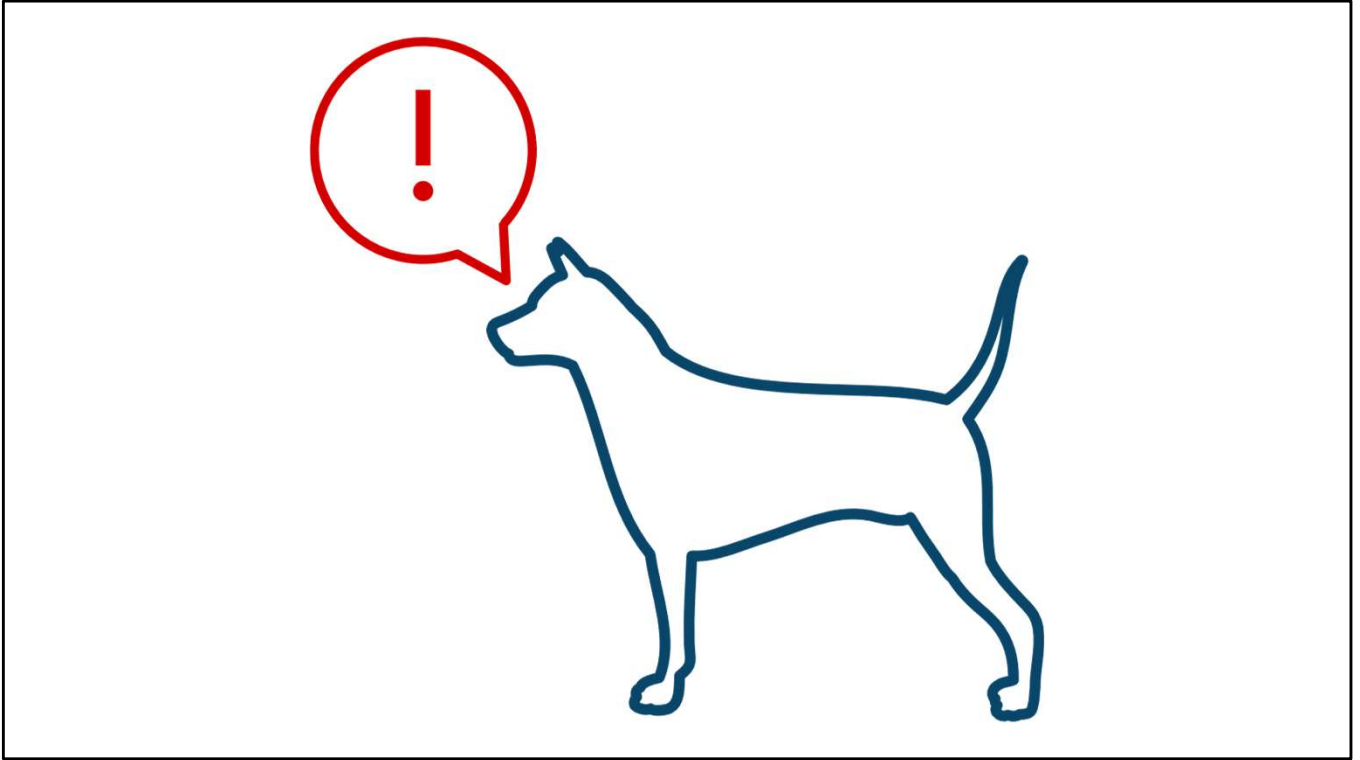
But, you notice we've got two aliases there that are actually going to interrupt a person? WakeUpMyBossAt4amOnSunday and TellMeIfThisHappensOneThousandTimesInARow?

Now we're getting into monitoring. Monitoring, in a nutshell, is about selecting a particular metric, establishing a threshold - what does normal look like in this context? - and then taking some kind of action if that threshold is exceeded.

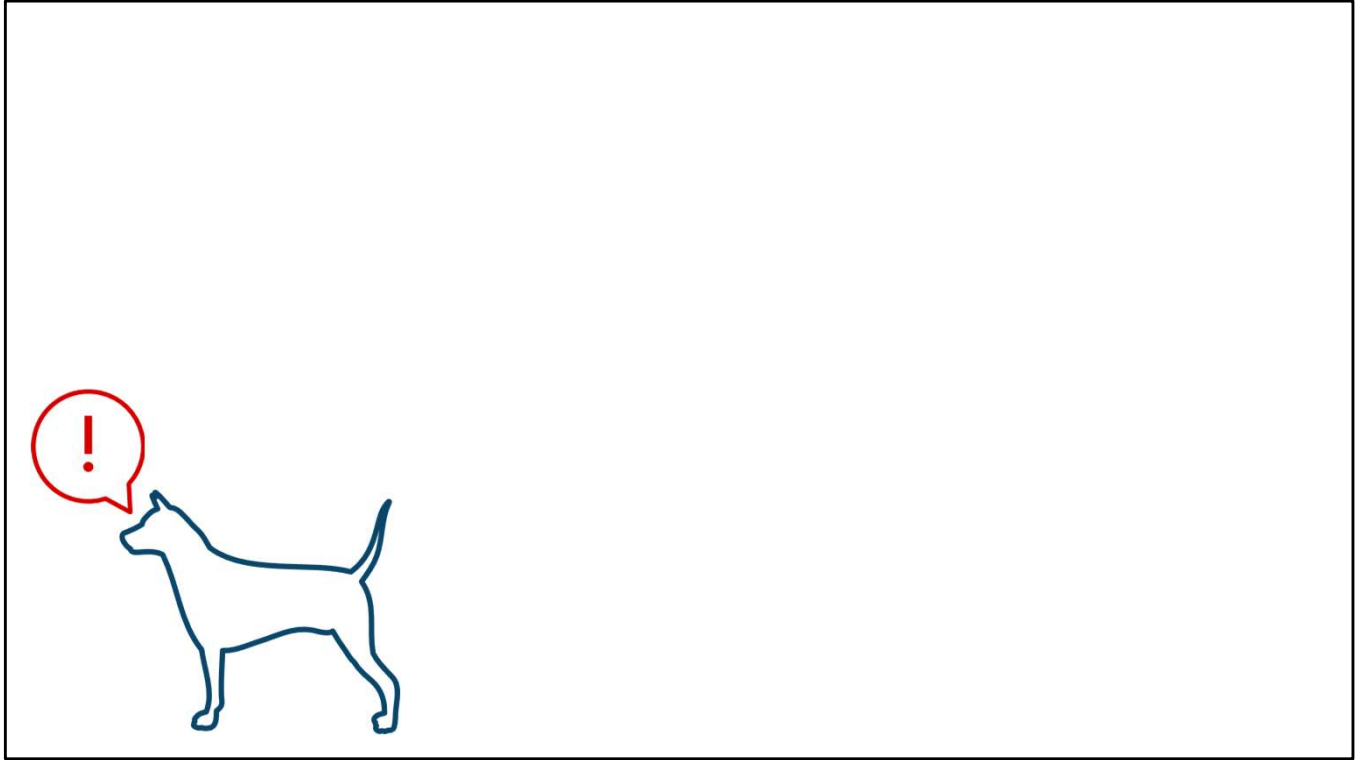


Let's go back to Sherlock Holmes. The title of this talk, the Curious Incident of the Log in the Downtime, is a reference to a detail from one of Sherlock Holmes' stories... The Adventure of Silver Blaze, and it's all about a racehorse that disappears the night before an important race, and the horse's trainer, John Straker, is found dead nearby, apparently beaten to death with a blunt instrument.

But what makes Silver Blaze interesting to us is that it features the closest thing the Victorians had to a monitoring system: a dog.

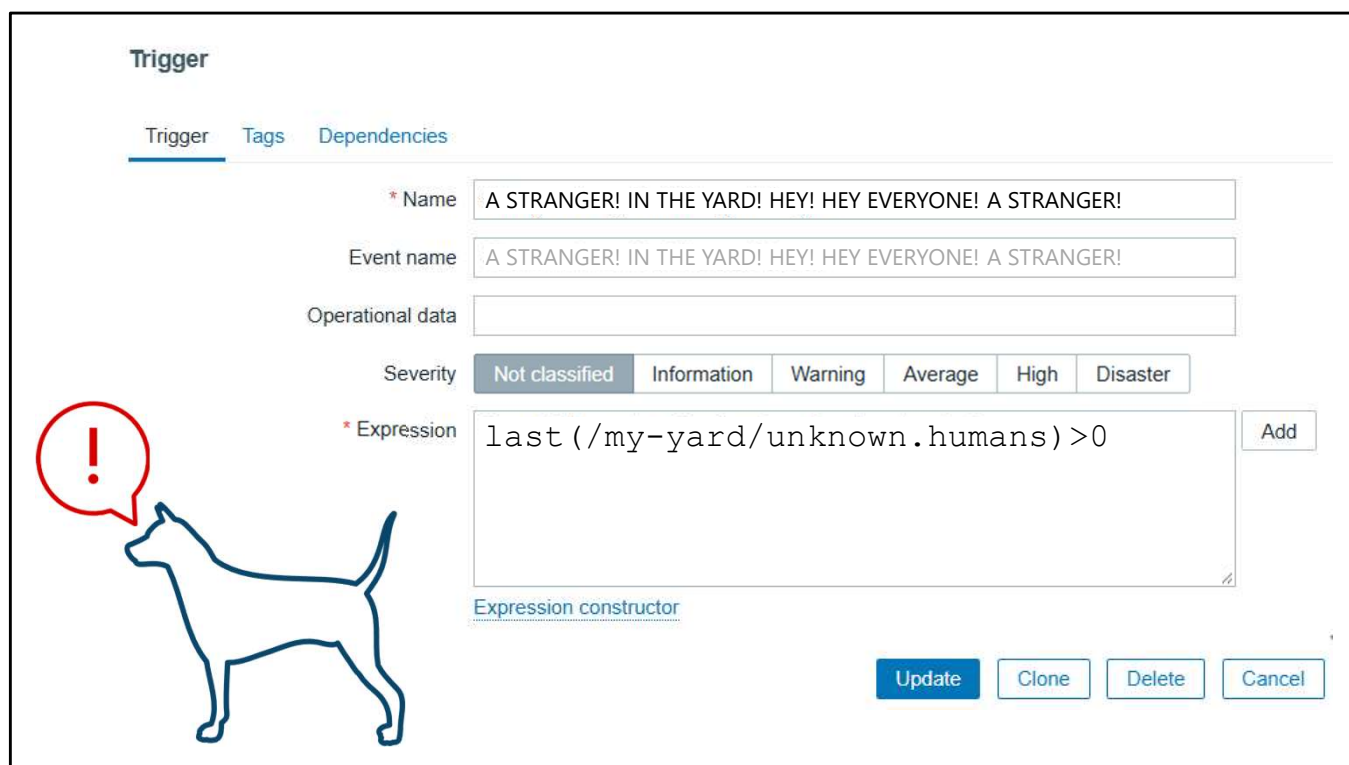


Let's configure the dog.



We're going to add a trigger here.

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Trigger

Trigger Tags Dependencies

* Name A STRANGER! IN THE YARD! HEY! HEY EVERYONE! A STRANGER!

Event name A STRANGER! IN THE YARD! HEY! HEY EVERYONE! A STRANGER!

Operational data

Severity Not classified Information Warning Average High Disaster

* Expression last (/my-yard/unknown.humans) > 0 Add

Expression constructor

Update Clone Delete Cancel

We'll give it a name - A STRANGER! IN THE YARD! HEY! HEY EVERYONE! A STRANGER! - and then set the alert expression: if the last count of unknown humans in the yard is greater than zero, sound the alarm. And, of course, we need to specify a severity: a stranger in the yard is definitely a disaster, right?

Looks pretty good, right? Except the expression here is based on "unknown humans". We're relying on the dog to decide who's a friend and who's a stranger... you ever lived next door to a dog that still barks at you even after you'd been there for years?



FALSE POSITIVE
(alert when there is no problem)



FALSE NEGATIVE
(problem but there is no alert)

That's a false positive: an alert when there is no problem. And if you've ever had a so-called "guard dog" that would happily make friends with a burglar in exchange for a snack? That's a false negative: a problem, but there's no alert.

"To the curious incident of the dog in the night-time."

"The dog did nothing in the night-time."

"That was the curious incident," remarked Sherlock Holmes.

The Colonel was very angry. "I have been on the turf for twenty years, and never was asked such a question as that before," said he. "A child would know Silver Blaze with his white forehead and his mottled off fore leg."

"How is the betting?"

"Well, that is the curious part of it. You could have got fifteen to one yesterday, but the price has become shorter and shorter, until you can hardly get three to one now."

"Hum!" said Holmes. "Somebody knows something, that is clear!"

As the drag drew up in the inclosure near the grand stand, I glanced at the card to see

they came up the straight. The six horses were so close together that a carpet could have covered them, but half way up the yellow of the Mapleton stable showed to the front. Before they reached us, however, Desborough's bolt was shot, and the Colonel's horse, coming away with a rush, passed the post a good six lengths before its rival, the Duke of Balmoral's Iris making a bad third.

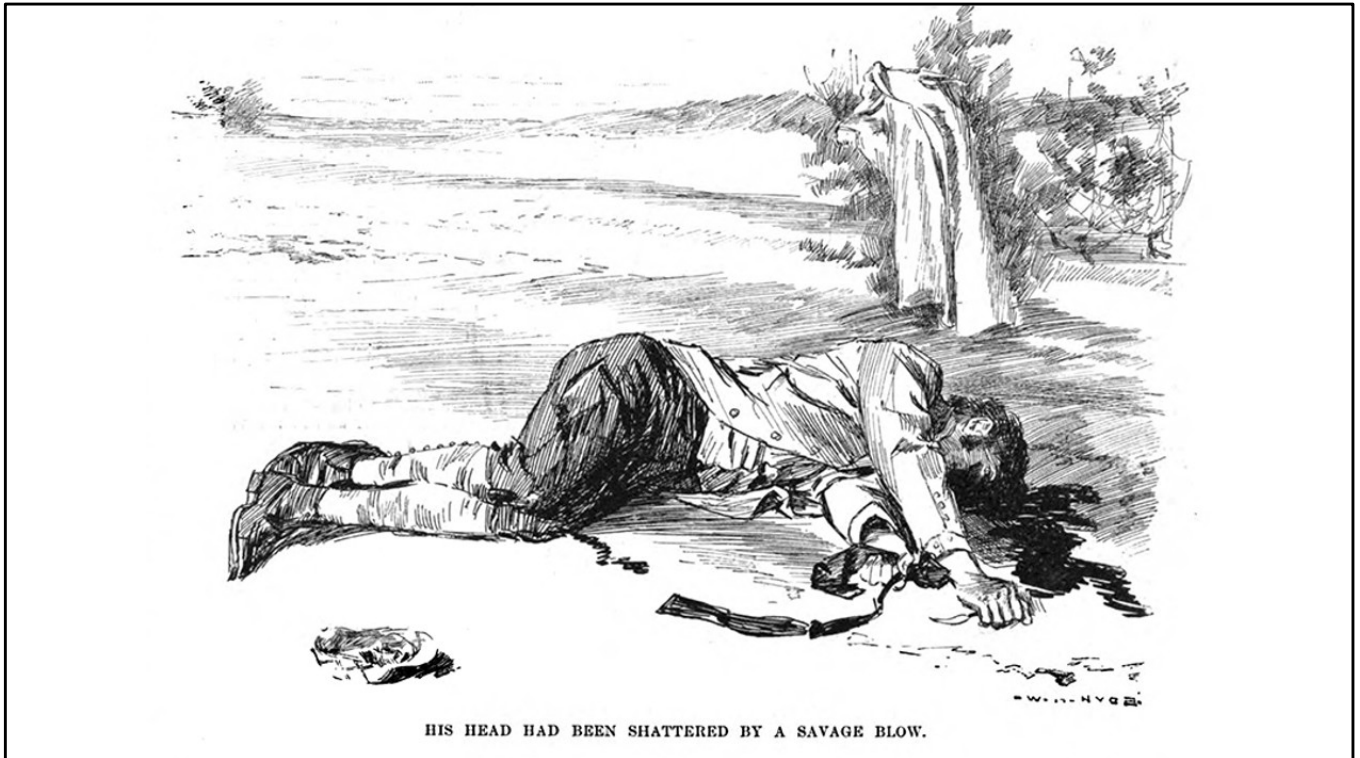
"It's my race anyhow," gasped the Colonel, passing his hand over his eyes. "I confess that I can make neither head nor tail of it. Don't you think that you have kept up your mystery long enough, Mr. Holmes?"

"Certainly, Colonel. You shall know

And in one famous scene from the Silver Blaze, a local policeman asks Sherlock if he's noticed anything interesting, and Sherlock draws his attention "to the curious incident of the dog in the night-time". "The dog did nothing in the night-time", replied his companion...

"That was the curious incident", remarked Sherlock Holmes.

In other words - the dog didn't bark - and so, assuming the trigger was configured correctly, whoever took the horse wasn't a stranger. The thief was somebody the dog recognised, and on a remote farm on Dartmoor, that leads to a very narrow list of suspects...



and sure enough, it turns out that Silver Blaze was actually stolen by his own trainer, John Straker, as part of a cunning plot to fix the outcome of a horse race - and in a particularly singular twist, the murderer turns out to be.... Silver Blaze; Straker wasn't attacked with a blunt instrument, he died from being kicked in the head by the horse he was trying to steal.



So that's the state of the art in systems monitoring in Victorian England: handwritten business ledgers and guard dogs. Logs and monitors. On-site, on-device - and if your office burns down? Well, good luck checking the log files to see if you'd just taken delivery of a supply of dangerous chemicals or anything... 'cos if the building burns, the records burn too.

Alright, we're going to stay in London - 'cos London's awesome, that's why I live there - but we're going to skip forward a few decades, to Agatha Christie's Poirot.



Let's skip forward a few decades, to London in the 1930s, and meet Agatha Christie's Poirot, the diminutive Belgian detective, known for his impeccable manners and immaculately waxed moustache - and, this being 1935, the police have a few tricks up their sleeve which weren't available to Sherlock Holmes:



(In this clip from Agatha Christie's Poirot, we see Chief Inspector Japp of Scotland Yard coordinating surveillance of a suspect using two-way radios to communicate with police cars, and a situation room with model cars being moved around on a giant map of London.)



That's right: motor cars, radios, and a big map. OK, the big map isn't actually a new invention - but when you combine the big map with a bunch of wooden cars and some long sticks, you get one of the earliest examples of a real-time visualisation.

But the biggest innovation here is the radio, because radio means remote units can send information to a central point where that information can be processed - and that means that we have telemetry.

tele + metry

telephone
television
telescope

meter
metric

"over a distance"

measurement

Tele, meaning "over a distance" - the same as in telephone, television, telescope - and metry, from the same root as metrics: measurements. Observations. Signals. And the absolutely key capability here is the ability to gather signals from remote systems and process that data centrally.



Agatha Christie's Poirot "The Lost Mine" © 1990 London Weekend Television Ltd

In the Poirot clip there, the police are using telemetry data to inform operational decisions: they're deciding what to do next based on a live data feed from their production environment, and allocating resources based on where they think those resources will be needed - and that's the power of telemetry in a nutshell; the people making the decisions don't need to be in the same place as the incident... which is also the basis for one of the great detective fiction double acts:



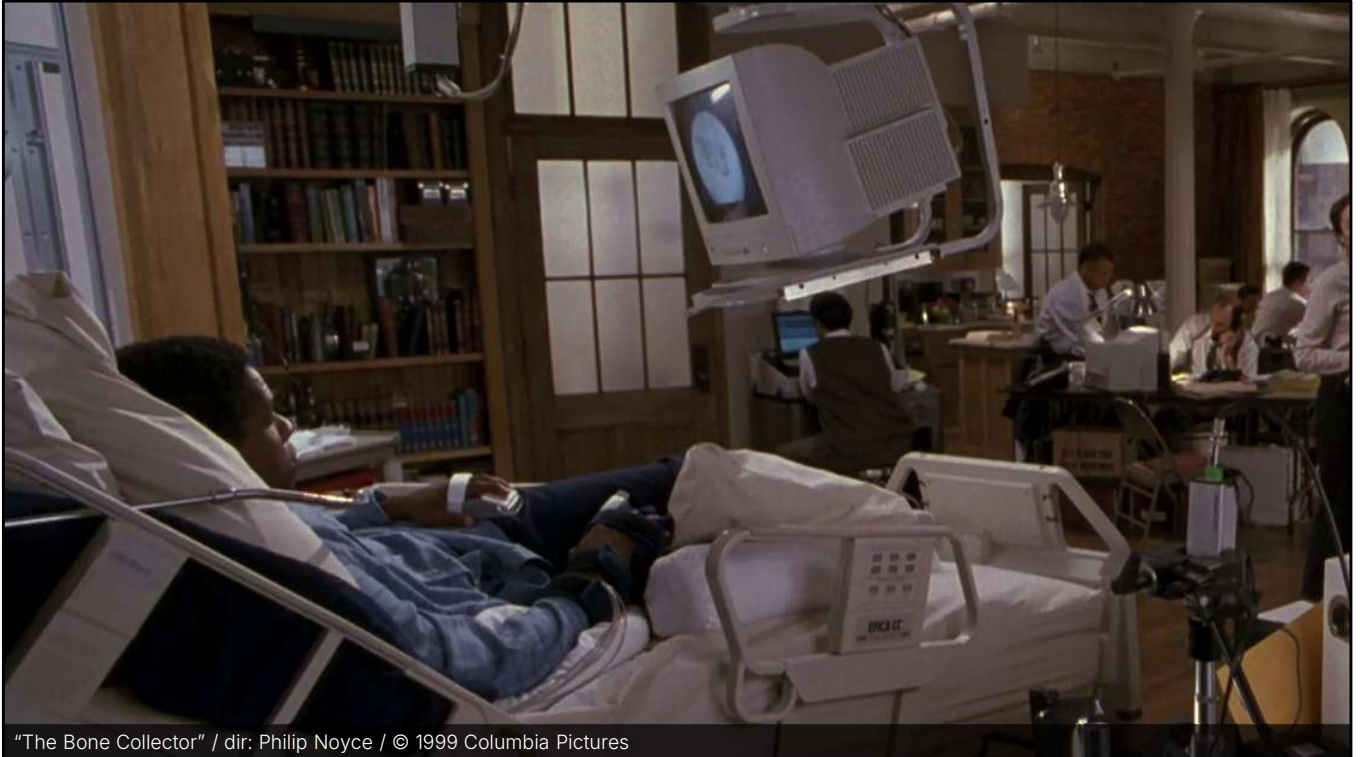
Lincoln Rhyme and Amelia Sachs, the protagonists of a series of bestselling novels by Jeffery Deaver. The first one, *The Bone Collector*, was made into a movie in 1997 starring Denzel Washington and Angelina Jolie - although Jolie's character was renamed from Amelia Sachs to Amelia Donaghy, 'cos apparently the filmmakers thought "sachs" sounded too much like "sex"... Hollywood's weird like that.



Lincoln Rhyme - Link - is a brilliant forensic investigator, but he's been paralysed from the neck down since his C4 vertebra was crushed in an accident at a crime scene fifteen years earlier. Amelia is a New York beat cop; she doesn't have the training or the experience to interpret the clues the way Link does, but she's observant, she's got a keen eye for detail and noticing things that are out of place... and so the novels are a case study in telemetry-based detective work.



Amelia's out there documenting the crime scenes, making notes, taking
photographs, and keeping in touch with Link via radio,



while he's confined to the hospital bed in his apartment, giving advice and analysis and trying to figure out what it all means.



Oh, and one tiny detail I thought was really interesting... that scene from Poirot? Film cameras and two-way radios: that was absolutely state of the art... and six decades later, the NYPD is still using film cameras and two-way radios, 'cos the year that movie's set, 1997, is like the absolute last moment before digital cameras and cellphones went mainstream; ten years later Apple brought out the first iPhone and as the saying goes, the rest is history...

history. There's an interesting word. Real-time telemetry is all very well, but what if you need to look something up **afterwards** - and it wasn't in the log files? It turns out there's also a whole load of scenarios where it's useful being able to look back at historical telemetry and ask: what else was going on when something interesting happened?

Let's meet another one of fiction's greatest detectives: Lieutenant Columbo.



What made Columbo such a brilliant show was that it completely turned the conventional detective story on its head. It wasn't a whodunnit - we knew exactly whodunnit, because every episode opened by showing us exactly who the killer was, and exactly how they killed their unsuspecting victim. The killers, and their victims, were usually movie stars, musicians, bankers, the high society of 1970s Hollywood, and the joy of the show is in watching Columbo, a down-at-heel homicide detective in a battered raincoat, put the clues together and figure out what the audience already knows.

Columbo's genius was at noticing details which didn't add up - and when you re-watch those old episodes, like I've done over the last few months as I was preparing this talk - all purely for research purposes, you understand - it's fascinating how often some kind of metric or measurement is the detail that unravels the killers alibi.



who can't possibly have killed his mother-in-law because he's got a rock solid alibi... see, Van Wick is an electronics genius, his house is fitted with all the most advanced home automation gadgets you can get in 1975 - stair lifts, automatic door openers, and even security cameras - and there's video footage of the murder happening, at the exact moment that Harold is across town at the opening of an art gallery.

Except... our eagle-eyed detective notices something:



A clip from Columbo series 4, episode 5, “Playback”. Columbo draws the suspect’s attention to a discrepancy in the video tape recordings of the evening of the murder. At the point the murder was apparently committed, the suspect was at an art gallery across town – apparently a solid alibi. But Columbo notices that in the video recording of the murder, Van Wick’s invitation to the gallery is visible on the table. The recording must have been created earlier, before Van Wick left for the gallery, and then replayed using a timer, making the security guards THINK that the murder happened while he was out.



The killer has edited the videotape... at the point he actually commits the murder, the security guards are watching a pre-recorded tape of an empty room, and then later, when he's downtown at the art gallery, schmoozing with several dozen witnesses, he uses a timer to play back the video recording of the murder... forgetting that he'd left his invitation to the art gallery on the table, in full view of the camera.

Dylan Beattie: The Curious Incident of the Log in the Downtime.

```
2026-09-21 09:14:02,117 INFO web.LoginController - Login attempt for user 'jsmith' from 10.4.12.88
2026-09-21 09:14:02,241 FATAL auth.PasswordValidator - Password mismatch for user 'jsmith'. CANNOT CONTINUE.
2026-09-21 09:14:02,242 FATAL auth.PasswordValidator - Hash mismatch (user typed 'Summer2026')
2026-09-21 09:14:02,243 INFO web.LoginController - Redirecting 'jsmith' to /login?error=1
2026-09-21 09:14:05,880 INFO web.LoginController - Login attempt for user 'jsmith' from 10.4.12.88
2026-09-21 09:14:05,991 INFO auth.PasswordValidator - Password OK for user 'jsmith' from 10.4.12.88
2026-09-21 09:14:05,994 WARN data.ConnectionPool - sudo escalation for user 'jsmith'
2026-09-21 09:14:05,995 WARN data.ConnectionPool - root database escalation for user 'jsmith'
2026-09-21 09:14:06,002 WARN data.NetworkMonitor - outbound data /eth0 exceeds configured limit (100Gb)
2026-09-21 09:14:06,002 WARN data.NetworkMonitor - outbound data /eth0 exceeds configured limit (200Gb)
2026-09-21 09:14:06,002 WARN data.NetworkMonitor - outbound data /eth0 exceeds configured limit (300Gb)
2026-09-21 09:14:11,302 INFO web.LoginController - session terminated for user 'jsmith'
2026-09-21 09:14:16,112 INFO cache.RedisCacheManager - Cache hit ratio 100% (0 requests served)
2026-09-21 09:14:16,113 DEBUG cache.RedisCacheManager - Returning stale value for key 'db-prod-01.status' = 'UP'
2026-09-21 09:14:18,540 ERROR email.NotificationService - Failed to notify on-call engineer: SMTP relay unavailable
2026-09-21 09:14:18,541 INFO email.NotificationService - Notification marked as delivered to preserve SLA metrics
2026-09-21 09:14:19,902 WARN auth.PasswordValidator - User 'mpatel' attempted login with Caps Lock off.
2026-09-21 09:14:19,903 FATAL auth.PasswordValidator - Account 'mpatel' locked. Reason: password correct
2026-09-21 09:14:21,377 ERROR jobs.NightlyBillingJob - Retry scheduled for 09:14:21,376 (1ms ago).
2026-09-21 09:14:23,004 WARN metrics.PrometheusExporter - Scrape endpoint returned 200 OK with body 'please stop asking'
2026-09-21 09:14:25,661 INFO health.HealthEndpoint - db-prod-01: DOWN, db-prod-02: DOWN, db-prod-03: DOES NOT EXIST
```

In information security, this is known as log tampering, and it's a common technique used in all sorts of target attacks and cybercrime. An attacker manages to get access to one of your systems - maybe they've exploited a vulnerability in your platform, maybe one of your users has reused a password. Point is, the attacker is in, but they've left a trail - entry point, IP address, maybe even records of which commands they ran and which files and resources they were able to access.

Dylan Beattie: The Curious Incident of the Log in the Downtime.

```
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2026-09-21 09:14:25,661 INFO health.HealthEndpoint - db-prod-01: DOWN, db-prod-02: DOWN, db-prod-03: DOES NOT EXIST
```

But, if they've got administrator-level access, a competent attacker will be able to cover their tracks - whether that's just deleting entire log files, or meticulously rewriting timestamps and modifying individual log entries so there's no trace they were ever in the system - just like our killer using pre-recorded videotape to rewrite the timeline of events.

Dylan Beattie: The Curious Incident of the Log in the Downtime.

```
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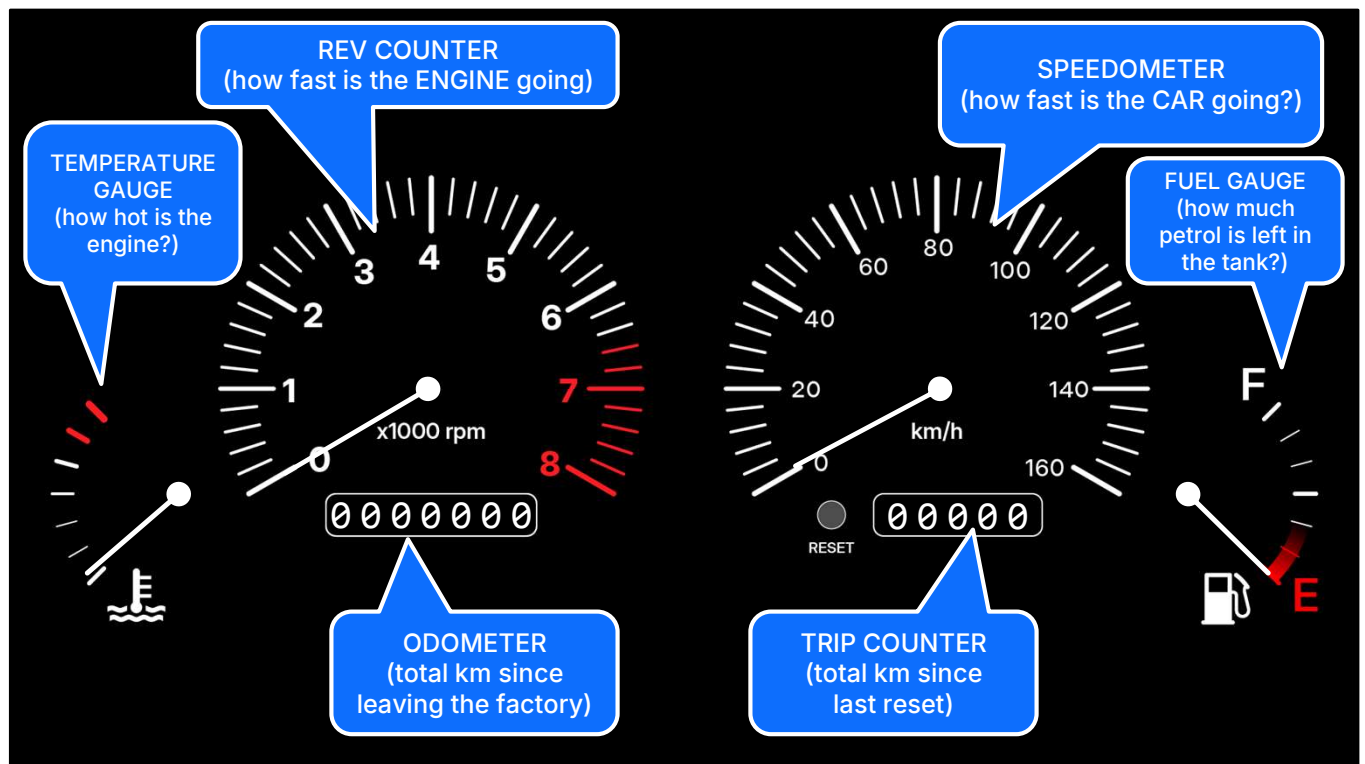
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2026-09-21 09:14:25,661 INFO health.HealthEndpoint - db-prod-01: DOWN, db-prod-02: DOWN, db-prod-03: DOES NOT EXIST
```

A recording doesn't need to be anything as complex as a videotape, though.



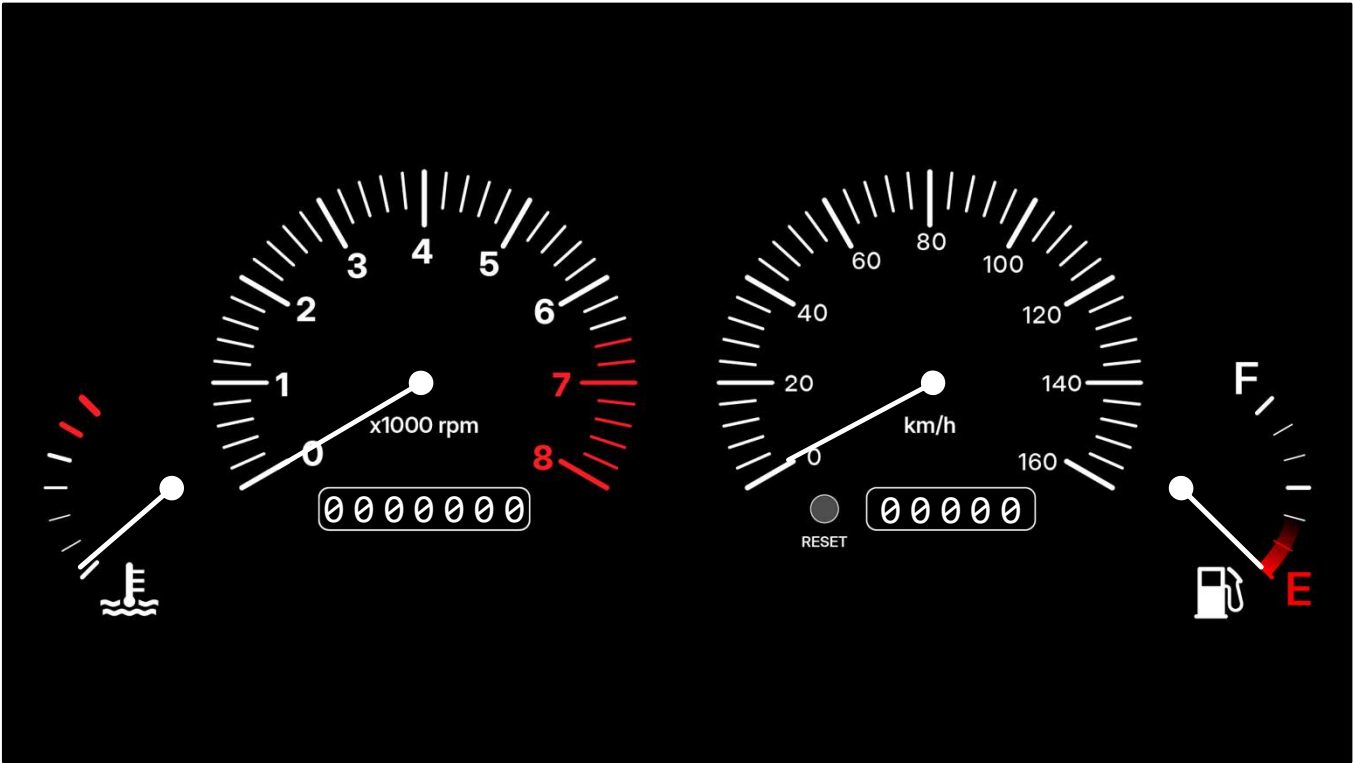
(A clip from Columbo S03E06 “Mind Over Mayhem” in which Columbo notices a discrepancy between a vehicle’s odometer and the mileage recorded in the log book.)

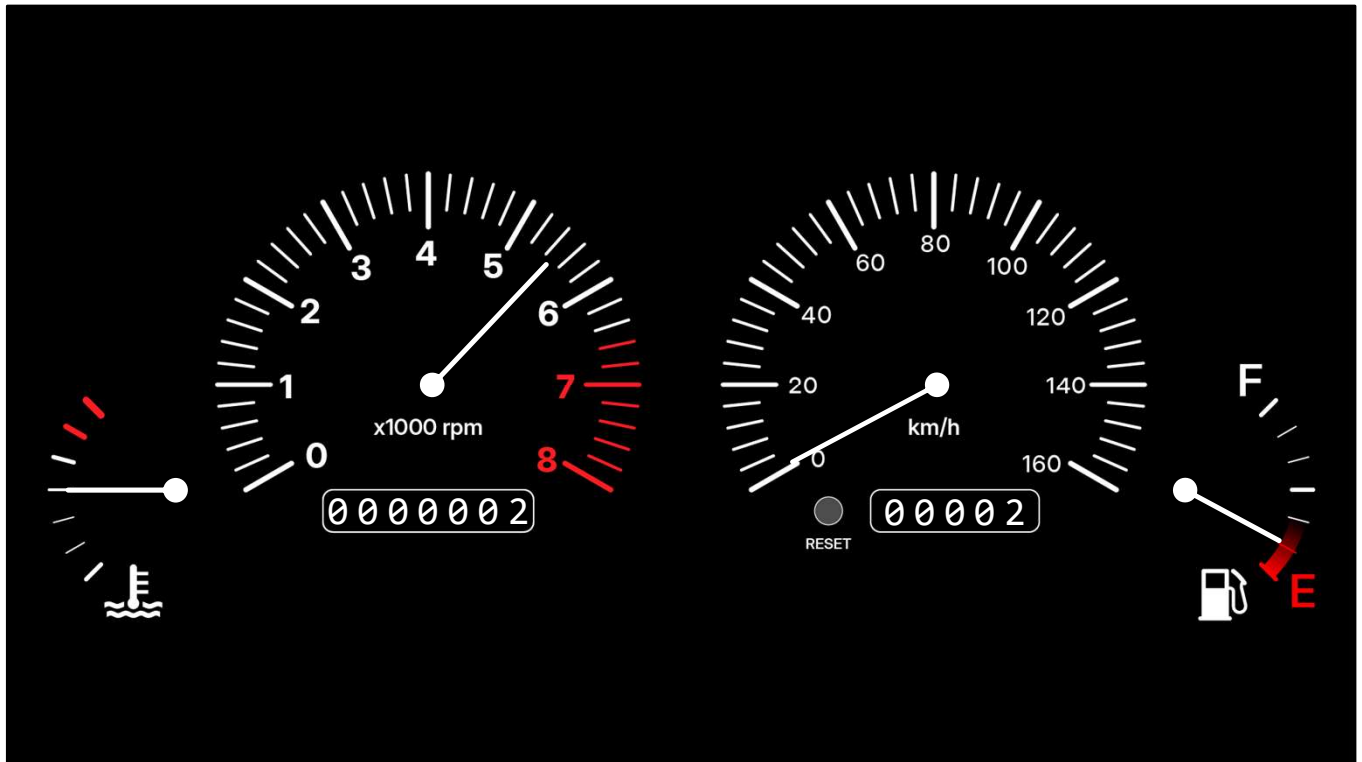
What Columbo noticed here is a discrepancy between two different sources - the log file, and the vehicle's odometer, the dashboard instrument that records how far the car has travelled. The thing about dashboards, though, is that all they do is show you the numbers. The art of systems management is about being able to interpret those numbers and figure out what's actually happening, just like Lincoln Rhyme solving crimes from his hospital bed - and turns out, you can figure out quite a lot just from looking at the instruments on a car dashboard.



So, here's our car. This is the the rev counter, or tachometer - how fast is the engine turning right now. Here's the speedometer - how fast is the car moving? We've got the odometer – how far this car has driven since leaving the factory – and the trip counter: how far since we last reset it. The temperature gauge – how hot is the engine? – and the fuel gauge.

OK, what can you tell me about this car?



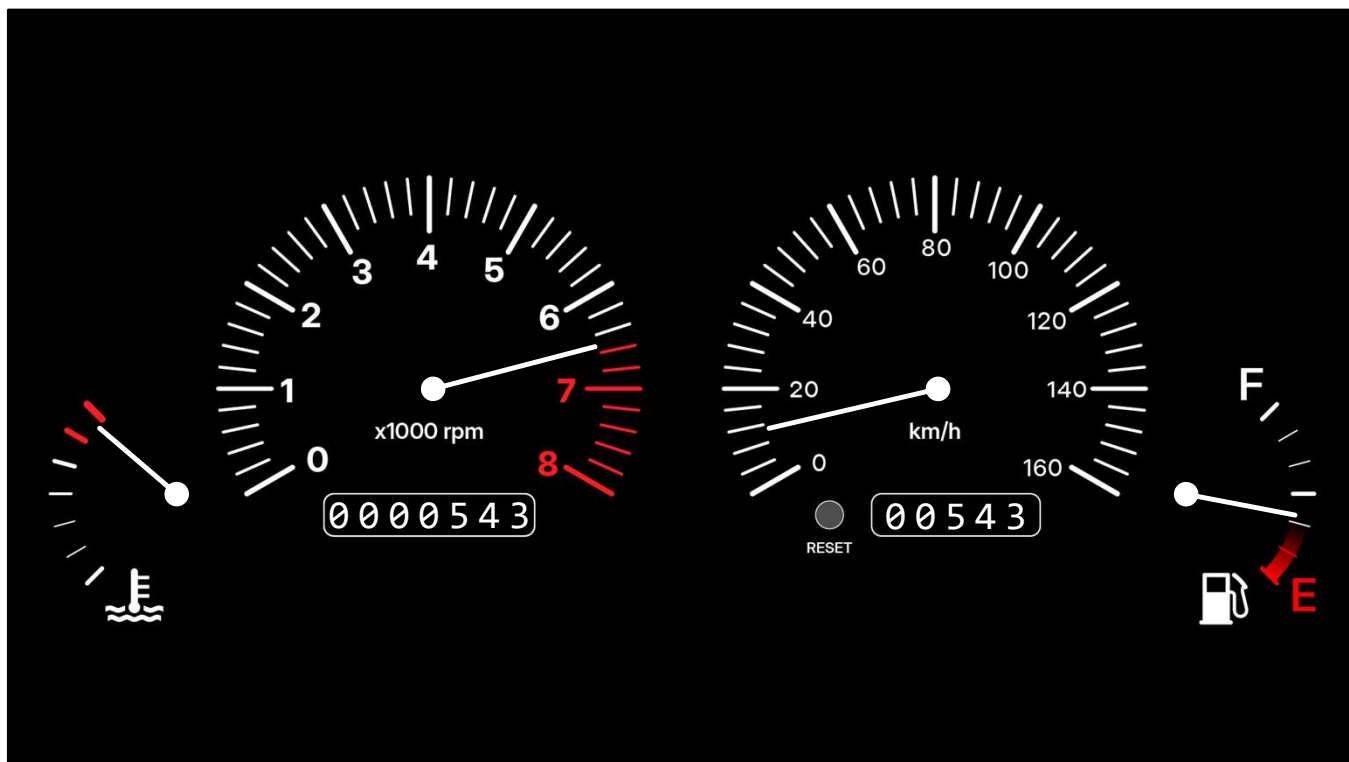


OK, so the odometer here is 2. This car has driven two kilometres - ever; the whole point of the odometer is that you can't reset it. It only ever goes up. It's got about a quarter of a tank of gas, and right now somebody's got their foot on the accelerator, 'cos they're revving the engine hard, but the speedometer reads zero so the car isn't actually moving... so what's the story?

Well, we can't be certain, but it's probably fair to assume that this car is brand new - it's only driven 2 kilometres, it's probably still at the dealer, and a potential customer is sitting behind the wheel revving the engine and grinning at the noise it makes. Or maybe a dealer mechanic is giving it a quick check before putting it into the showroom.

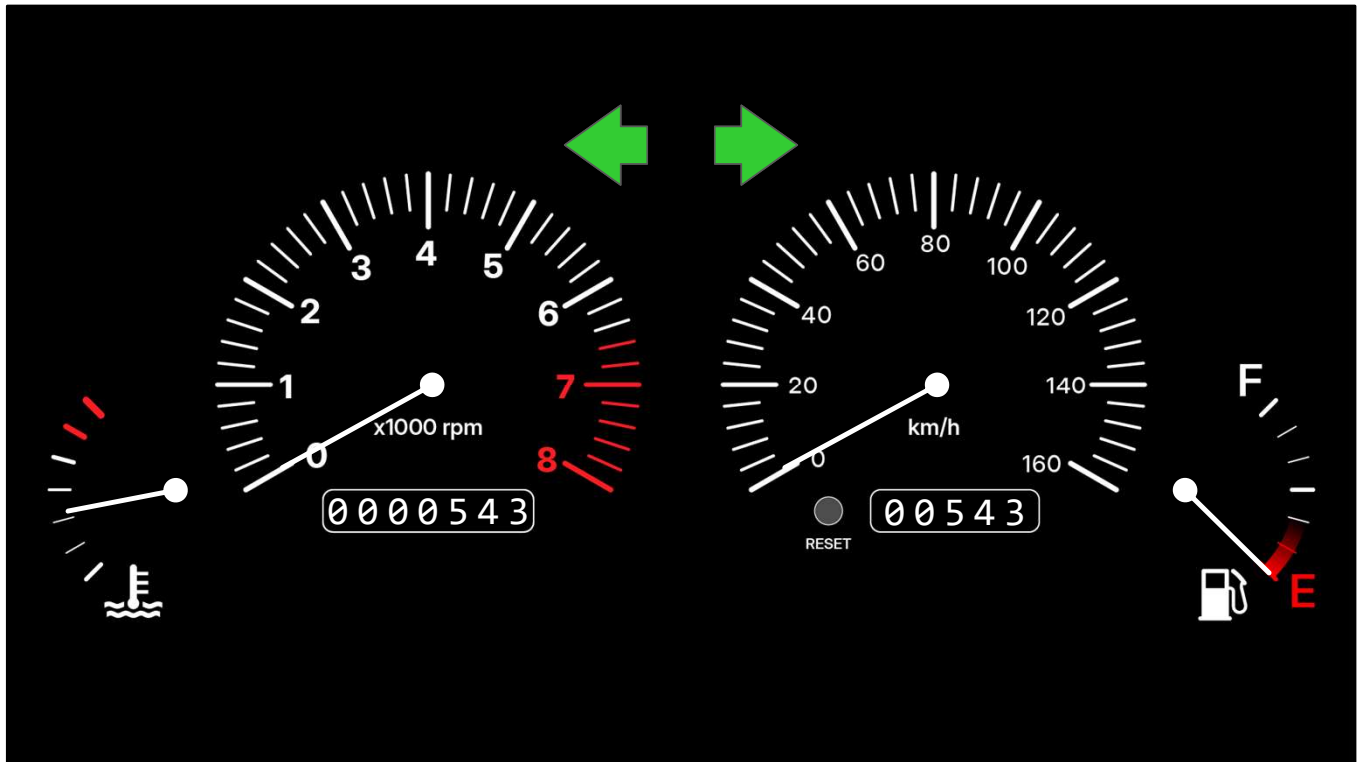
How about this car?

Dylan Beattie: The Curious Incident of the Log in the Downtime.



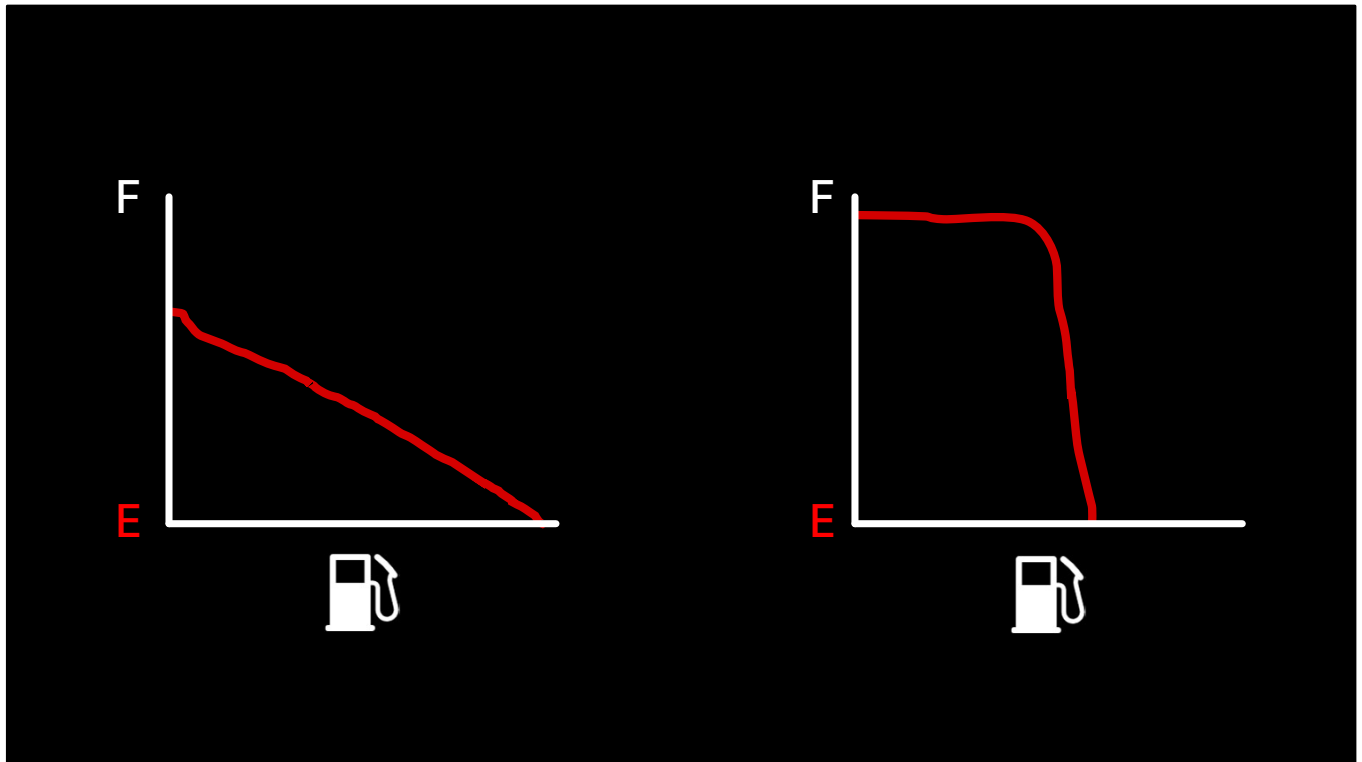
Somebody doesn't know how to drive; they're in way too low a gear, they're running the engine REALLY hard but they're only doing 10km/h, and you can see the engine temperature there is way up in the danger zone.

How about this one?



Yep. This car is stopped at the side of the road because it's run out of fuel - and you can see the blinking lights here; the driver's put on the hazard warning indicators 'cos they're probably stopped somewhere.

Can you tell from this whether there's a mechanical fault? Well, no - the car's stopped because it's run out of fuel. That's just a thing that cars do. But now let's imagine that the fuel gauge on this car isn't just a simple gauge, it's a log that shows fuel levels over time.



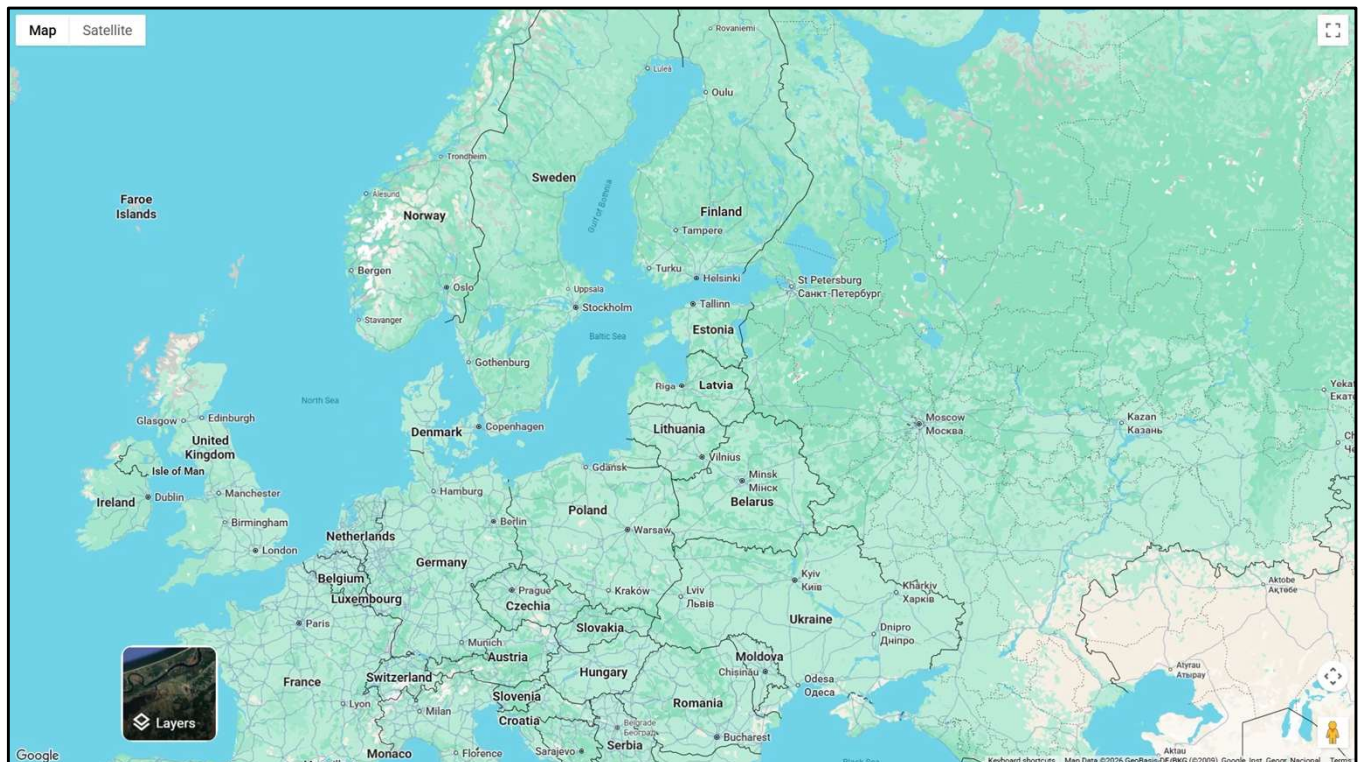
What's the difference between these two scenarios?

Yep. One of them, we just ran out of fuel 'cos we forgot to fill up the car. But the other one - something weird has happened; we've suddenly gone from having half a tank of petrol to having zero petrol. The METRIC is the same - no fuel - but the story behind it suggests two very different resolutions; one of them, we just need to fill the tank, the other one, we need to find a garage and figure out what's wrong with the fuel system.

At various points in my own career, I've had to deal with a production incident where a critical system - usually the main database server - stopped responding, and over time, I realised that just knowing that the database server has run out of memory isn't actually helpful: what's useful is knowing how memory usage has changed over time.

The point where things get really interesting, though, is when we start combining telemetry from multiple systems.

Dylan Beattie: The Curious Incident of the Log in the Downtime.



Any of you folks use Google Maps when you're driving? You know how in Google Maps you can turn on all sorts of different layers - and one of those layers is Traffic, that'll show you where the traffic jams are happening right now? You ever wonder where it gets that data from? Well, turns out that the whole time you're using Google Maps, your phone is sending telemetry data back to Google - where your phone is, and how fast it's moving. Google says it's all anonymous, and you can opt out if you want to, but most people don't bother - which means at any given point, Google has a huge amount of real-time data about how many phones are moving along a particular stretch of road, and how fast those phones are moving, and if there's a hundred phones all crawling along the same road at about five kilometres per hour, Google concludes that those devices are all stuck in the same traffic jam.

And, most of the time, that's a reasonable conclusion. Most of the time.

In February 2020, a guy called Simon Weckert rented a hundred Android phones, opened Google Maps on all of them, put them in a little red plastic trolley,

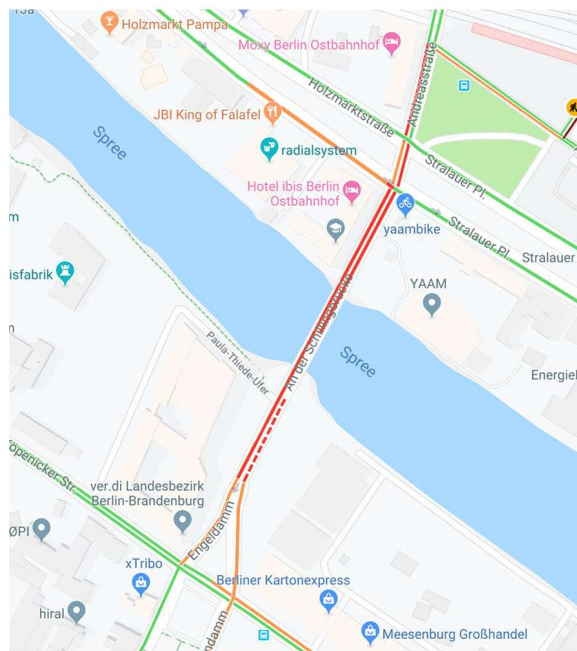
Dylan Beattie: The Curious Incident of the Log in the Downtime.



<https://simonweckert.com/googlemapshacks.html>

and pulled his trolley around the streets of Berlin to see if he could create fake traffic jams - and it worked. Within half an hour of Simon wheeling his little trolley down a given street,

Dylan Beattie: The Curious Incident of the Log in the Downtime.



<https://simonweckert.com/googlemapshacks.html>

Google Maps would show that street as being heavily congested with slow-moving traffic, and would start sending motorists on alternative routes to avoid the traffic jam; in a few instances, Simon creating a fake traffic jam on one street actually caused a real traffic jam a few streets over.

That's one very simple example of combining telemetry from multiple sources to produce actionable insights. One phone moving slowly? That's just a pedestrian. Four, five phones moving together slowly? Maybe a group of friends out for a walk... but fifty phones moving together slowly? Probably a traffic jam. Three hundred phones moving together at exactly two hundred kilometres an hour? That's probably an express train. One phone doing fix hundred kilometres an hour? Some idiot's on a plane and not switched their phone off.

But take a moment to think about a modern smartphone. GPS location data is just one of the signals that your smartphone captures. Cameras, accelerometer, microphones, compass, the ability to detect other nearby devices using Bluetooth and Wi-fi... one single smartphone can gather an astonishing amount of information, and since just about everybody carries a smartphone with them all the time now...



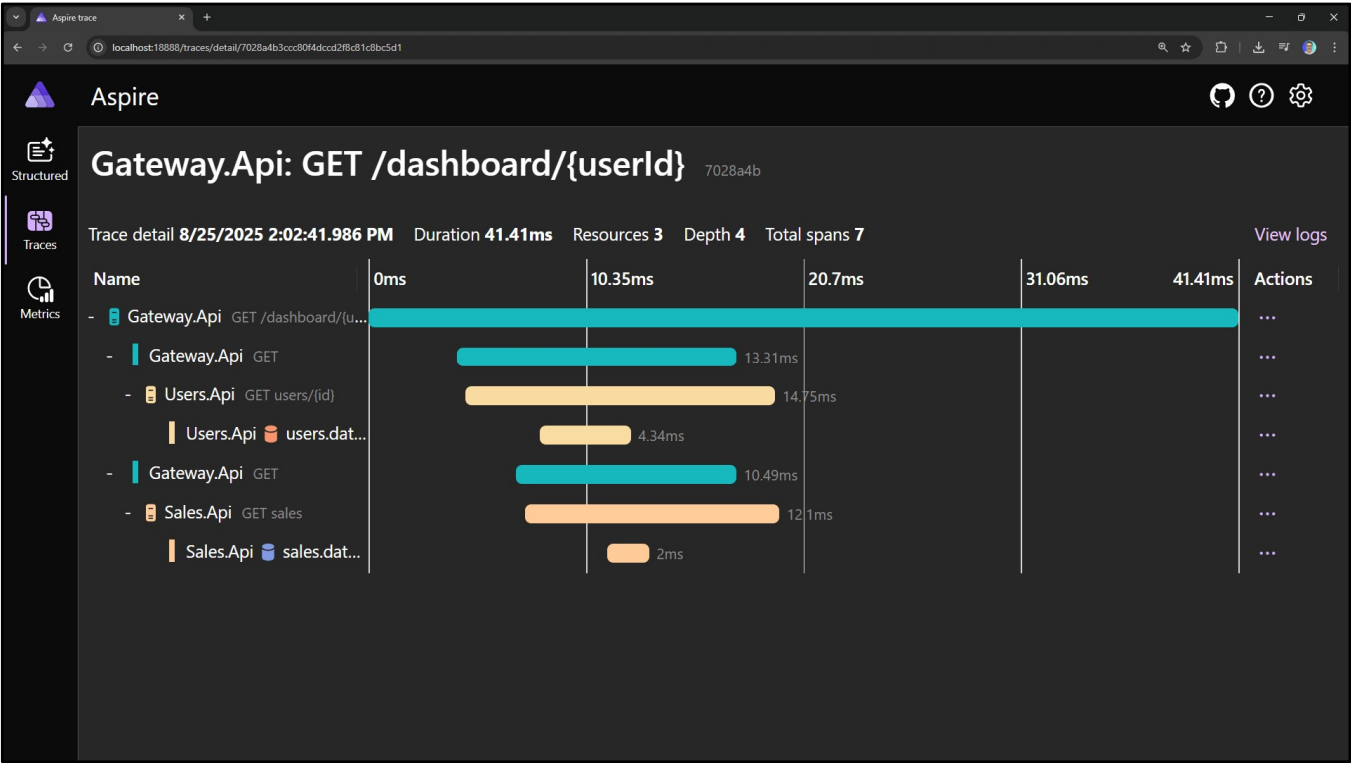
A scene from “The Dark Knight” in which Batman turns every mobile phone in Gotham into a node in a massive surveillance array.



Batman, the World's Greatest Detective. I always loved Batman 'cos he's a superhero with no actual superpowers - he's not an alien, he's not magic, he can't fly or read minds: he's just a guy who happens to be very clever, good at martial arts - and absolutely stinking rich. And in that scene from Christopher Nolan's "The Dark Knight", Batman turns every mobile phone in Gotham City into a listening node in a giant sonar array... he knows it's wrong, he knows it's immoral, unethical, illegal, and could be used for all sorts of evil deeds if the wrong people were in control of it. But with The Joker holding Gotham to ransom, threatening to kill hundreds of innocent people, Batman decides that the ends justify the means, and as a final additional safeguard, he designs the system so that the only person who can access it is his friend and confidante, Lucius Fox.

We'll come back to this in a moment, but first I want to talk about the other scenario where combining telemetry from multiple sources is useful is creating a connected narrative: the ability to track one single request or subject as it moves through a series of connected information systems.

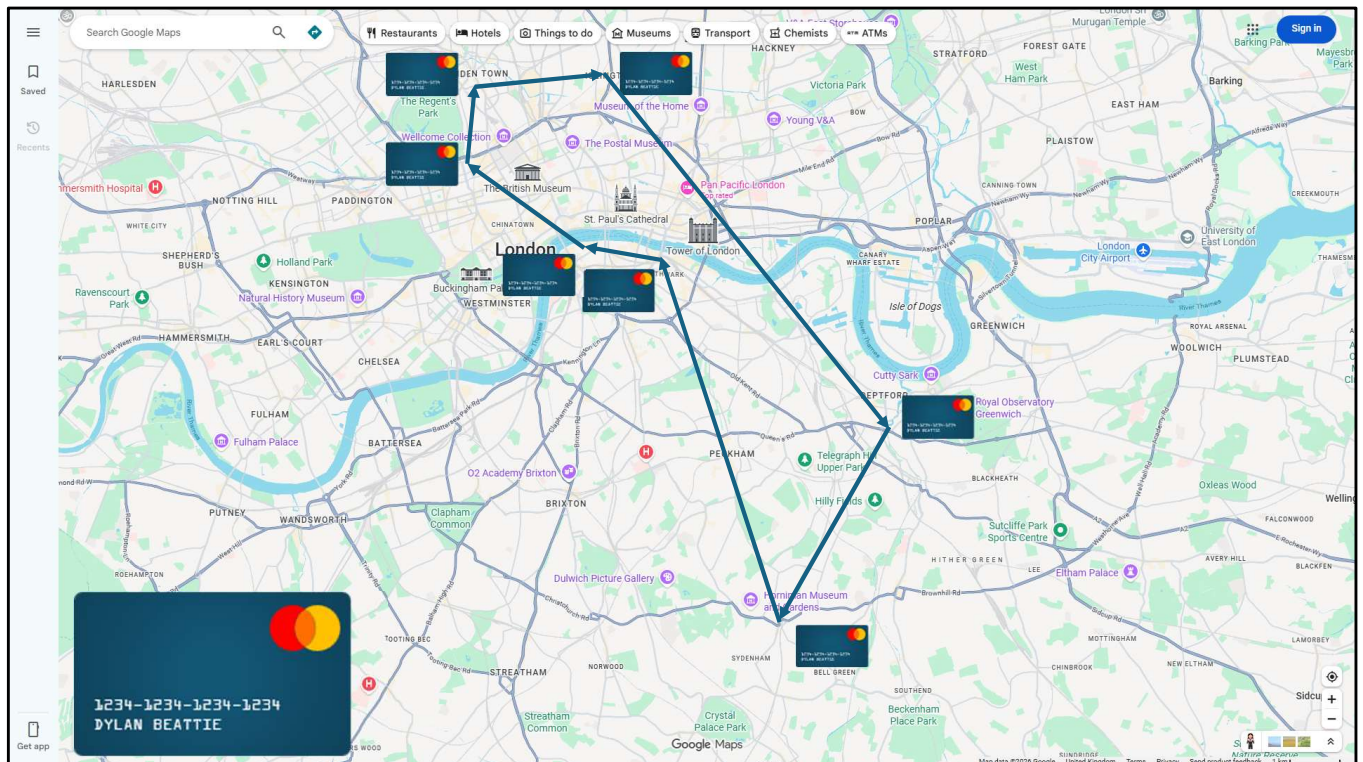
Dylan Beattie: The Curious Incident of the Log in the Downtime.



In OpenTelemetry, these are known as spans and traces - a span represents a single operation taking place within a single system, a trace is a series of spans joined together to create a full picture of what happened, and the thing that makes it all possible is something called a correlation ID.

Now, I tried really hard to find a good example from detective fiction of somebody solving a mystery using a correlation ID... but I couldn't find one. And I think the reason for that is that, when we apply these kinds of techniques to fields like law enforcement, they don't actually make for very interesting stories.

Dylan Beattie: The Curious Incident of the Log in the Downtime.



I live in London. Let's imagine a typical day out in the city. I'll walk the ten minutes or so from my house to the train station. All the trains in London have contactless payment now, so I use my card to touch in when I get on the train. Half an hour later, I get to London Bridge, I touch out - same card. I walk along the river, stop to get a cup of coffee - same card, same number. I find a cafe, stop for lunch, pay with the same card. After lunch, I get on a bus - yep, contactless payment, same card. I meet some friends in a bar, buy a round of drinks - same card. We go to see a movie, I use my card to buy the movie tickets, and then finally take a train home.

Now imagine that, for some reason, the Metropolitan Police are monitoring my movements. All they need to do is get access to all of those databases, and they can build a complete picture of exactly where I went, how I travelled, what I ate, what I drank... all because every one of those transactions has the same common identifier: the number on my bank card.

You can see why that wouldn't make a great detective story: it's just somebody sat in front of a terminal running database queries for an hour or two.



There are plenty of identifiers out there which law enforcement has used to track persons of interest across multiple systems. Your passport number, your bank card number, the license plate on your car... London has various restricted traffic zones these days, places where you have to pay to drive your car, and they're all monitored by cameras with automated number plate recognition; you go down the wrong street, or make an illegal turn, an automated camera snaps your license plate, looks up your address, and sends you a penalty notice - the first time a human gets involved is when you open your mail.

Up until now, though, there's always been a sort of implicit contract involved. Cars are useful and convenient, but they're also dangerous, and so I accept that if I want to drive around in my own car, I need to licence it and accept that my details are going to end up in a bunch of databases. Bank cards and contactless payment are wonderfully convenient, but again - I accept that they only work because somebody else is running a whole lot of expensive technology and infrastructure so that I can pay for my oat cappuccino by waving a card across a terminal, and I accept that those organisations need to protect the integrity of their systems by doing things like logging transactions and triggering alerts if something weird happens. And it's always felt to me like that's the deal: if you don't want to get tracked, take public transport and pay cash. 20 years ago, that was actually pretty easy.



Today... ok, question: who's paid for something today using cash? This week? This month? The scary fact is that we can't opt out any more. If you want to earn a living, pay your bills, pay your taxes, buy groceries, it's practically impossible to do that in most places now without leaving an identifiable electronic trail.

Life In 2026: Terms and Conditions

In order to provide essential services, your activities and communications will be tracked, monitored, watched, recorded, logged, analysed, sampled, registered, correlated, and stored in a hundred different databases which could be sold to anybody at any time.



I accept the terms and conditions

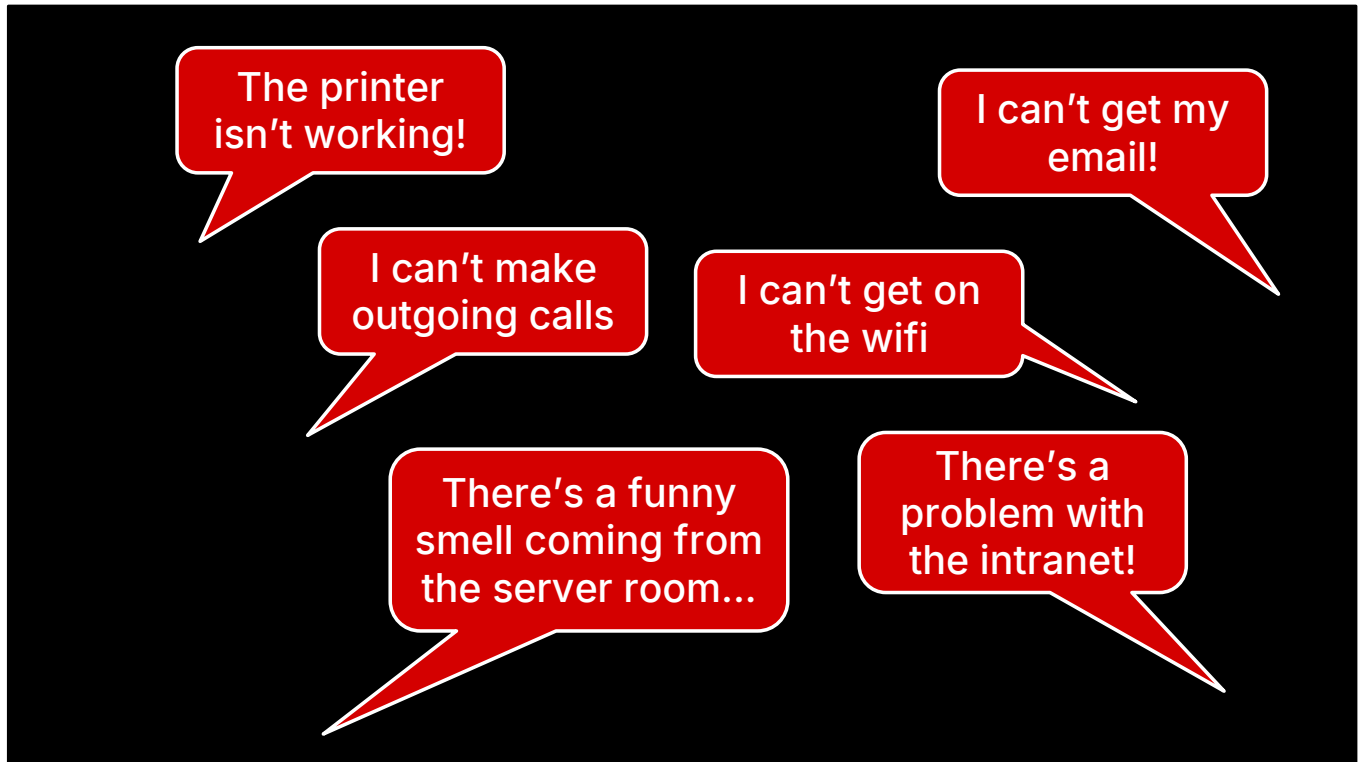
Error: You Must Accept the Terms and Con



And with recent advances in facial recognition, voice recognition, we're close to a point where just leaving your apartment will be enough to leave digital footprints in somebody else's database... and so you have to ask: who owns the database? Is it Batman? 'Cos Batman's database is null-key encrypted so that's probably fine... but what if it's NOT Batman?

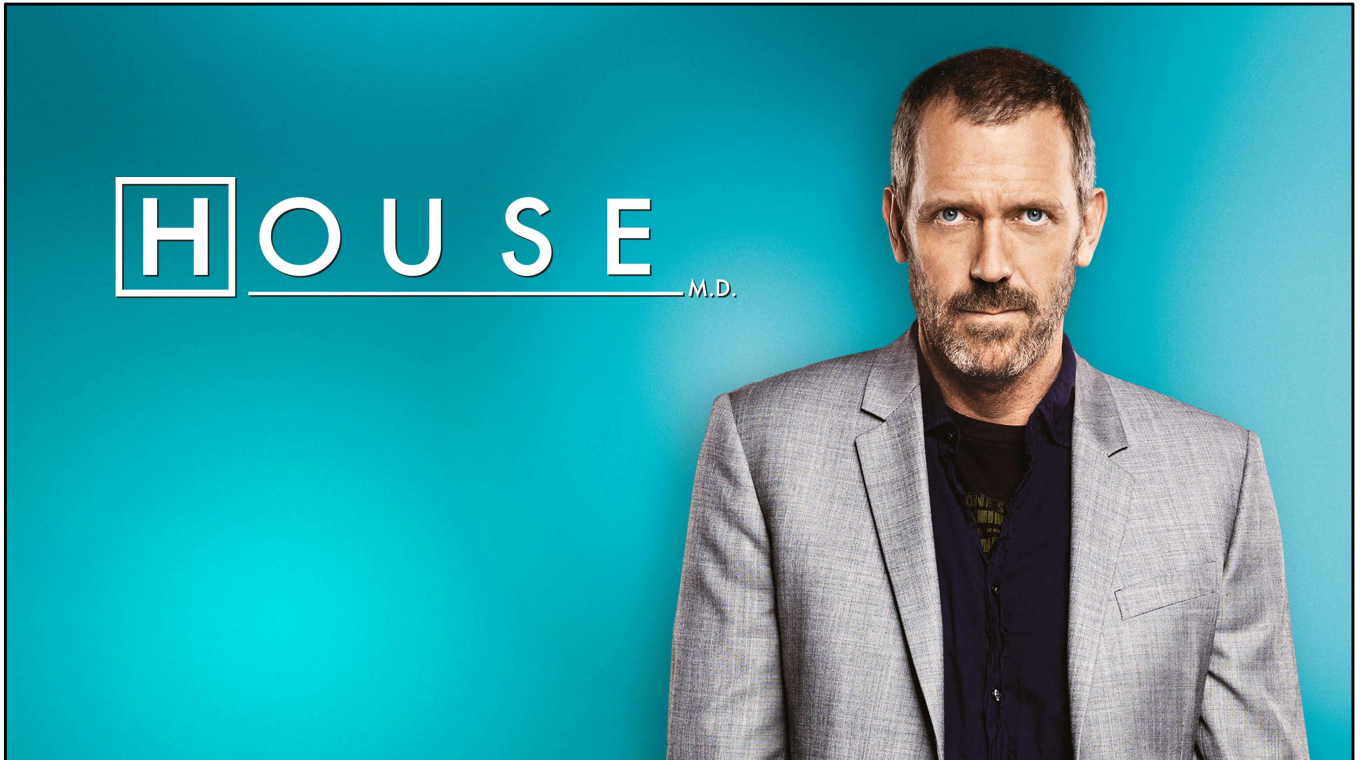


Philip K Dick's novel *A Scanner Darkly* is set in a dystopian future, where the protagonist wears a "scramble suit" so that cameras can't track him – this is from the 2006 movie adaptation starring Keanu Reeves, but the book was published in 1977... last year it was banned in... well, let's just say a European country that isn't exactly noted for its excellent track record on human rights. Apparently the book contains information "capable of harming the national interests of our country"... which, ok, is probably because everybody in the book is high on drugs pretty much all the time, but it does make you stop and think.



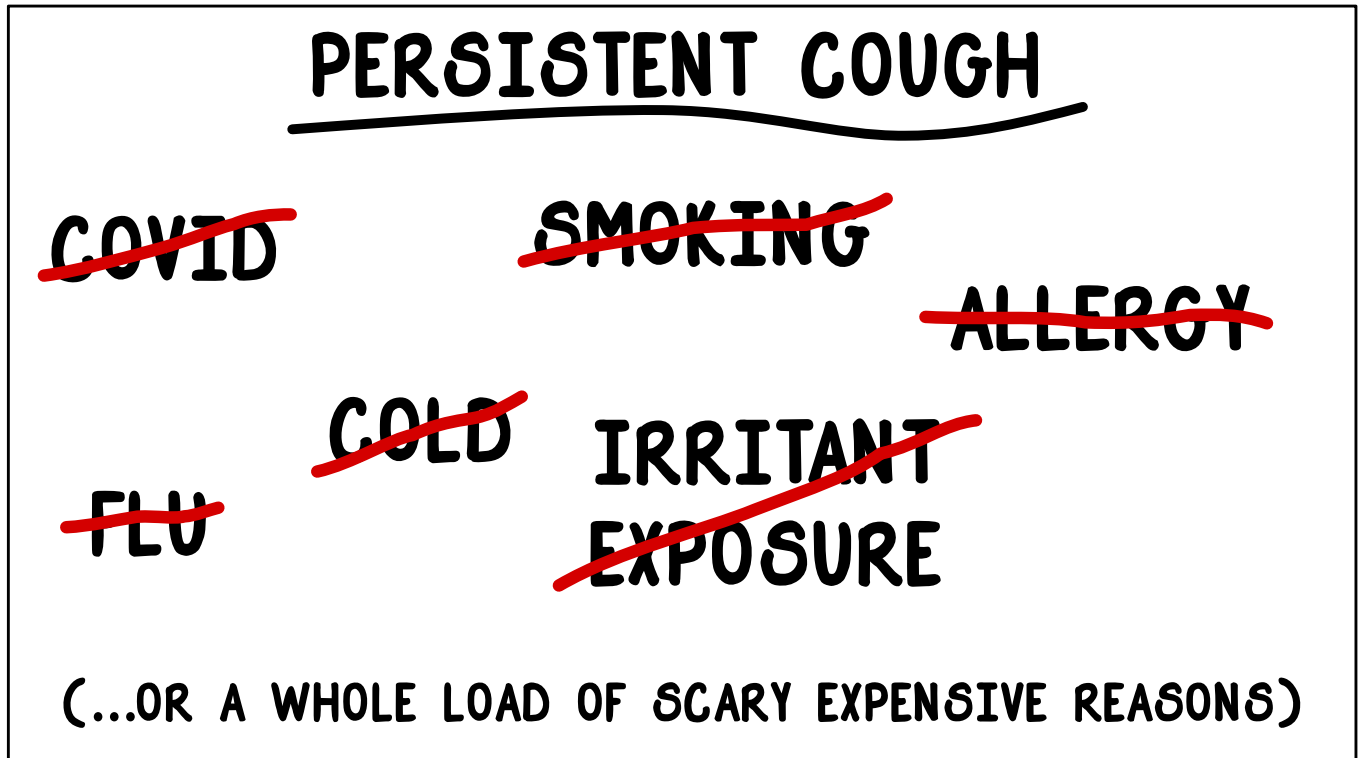
The last thing I want to talk about today is something that I'm sure many of you folks will be able to relate to... what happens when you've got multiple alerts that, at first glance, appear to be completely unrelated.

The definitive literature on this kind of scenario is another one of my favourite fictional detectives... somebody who, at a glance, you might not even realise is a detective at all. This guy.



Doctor Gregory House. I love this show - great writing, great characters, fantastic performances... if you've not seen the show, here's the premise: Dr. House is a complete jerk. He's rude, lazy, selfish, arrogant, narcissistic... problem is, he's also a medical genius; he's the guy the other doctors send their patients to when they can't figure out what's wrong. At the heart of the show is a technique known as differential diagnosis - which I'm sure a bunch of you folks out there have applied when you're troubleshooting software and network problems, even if you didn't realise that's what you're doing.

It works like this. Say a patient shows up at your clinic complaining that they've got a cough. That's their only symptom: a persistent, tickling cough. Now, there are a whole bunch of conditions that could explain that symptom, and so what you need to do is draw up a list of possible explanations, and then figure how to systematically eliminate them.



So... what could cause a cough? Could be COVID. Could be caused by smoking. Could be an allergic reaction, could be they've been exposed to some sort of irritant, could be a common cold, or the flu... or a whole host of really scary expensive things.

What you need to determine is how likely each explanation is, and how expensive it would be to eliminate them - whether that's the cost of a test, or the time taken to get a result, or the risk to the patient. COVID testing is easy and cheap, so start with that... OK, negative. Not COVID. Do they smoke? Never smoked? Probably not that then. Cold? Flu? No, it's been going on for a month now. Allergies? Take an antihistamine... didn't help? Have they been doing construction work? Sawdust, brick dust, varnish fumes... as long as your patient's telling the truth, you might be able to solve the mystery without running any tests at all... but no, it's not that.

And now we're down to the scary expensive things: x-rays, scans, biopsies - because whatever's happening here, it's not something doctors see every day, like a cold or an allergy; it's a medical mystery.

Let's do a differential diagnosis of a software application. A customer reports that they're not getting any of our emails. OK, list of potential explanations.

CUSTOMER ISN'T RECEIVING EMAILS

~~DID WE
ACTUALLY
SEND ANY?~~

~~ANTI-SPAM~~

~~SMTP
RELAY~~

~~JUNK
MAIL~~

~~WRONG
EMAIL~~

~~NETWORK~~

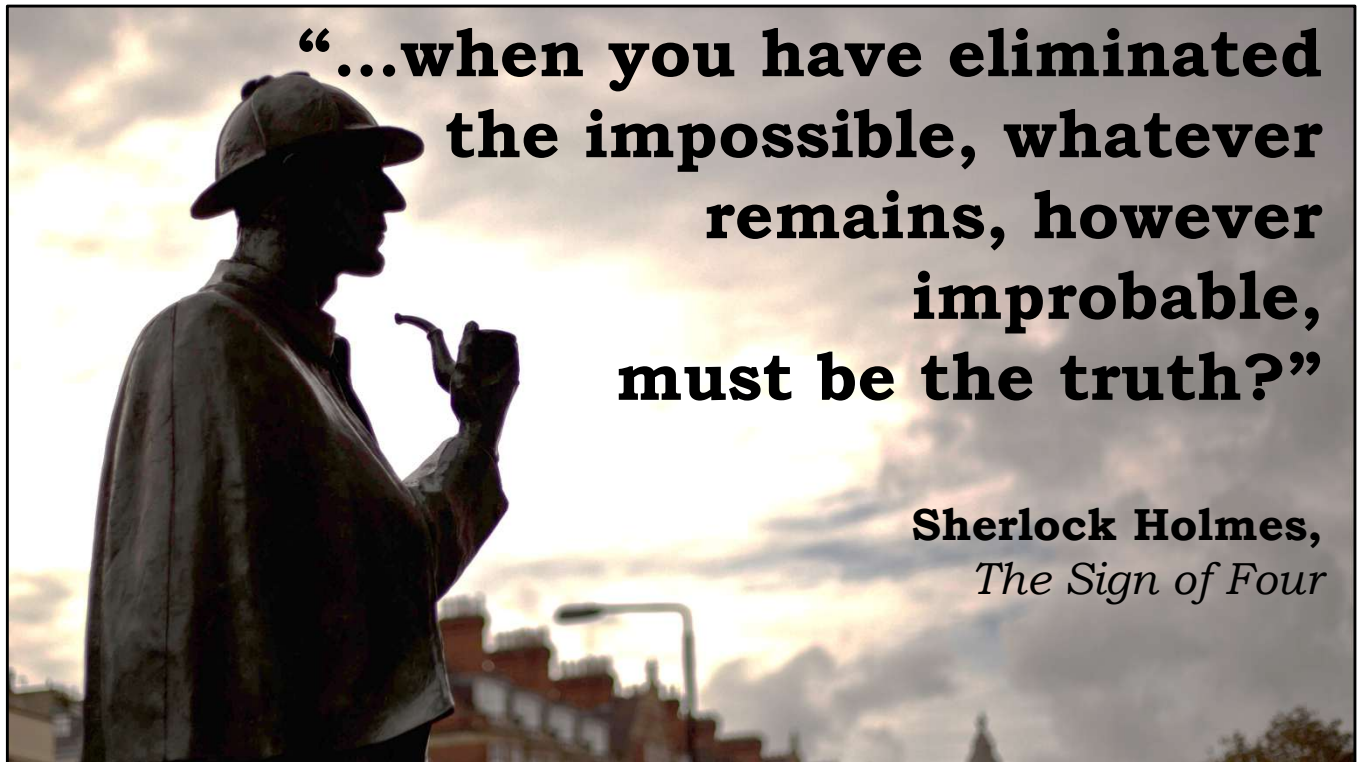
(...OR MAYBE IT'S DNS)

Maybe we didn't send them any email. Maybe there's a new anti-spam system, or a problem with the SMTP relay. Maybe they gave us the wrong email address – you'd be astonished how many people don't know their own email address. Maybe it's a network problem. Maybe it just went to junk mail.

Or maybe it's DNS

OK, let's eliminate some things. First – check we've got the right email address. Actually send them something. It arrived? Not that, then. Did it go to junk mail? Nope. Did we actually send them anything? Check the logs. Is it a network issue? Ping some stuff. Network's fine. SMTP relay? Can we send other emails through it? Do other customers get email? They do. Not the SMTP relay, then.

Has their ISP installed some sort of new anti-spam system? Well, that's probably harder to check... but we can find out. Maybe it's MIMecast or something blocking our messages. But if it's not? It's probably DNS. It's often DNS.



There's a famous quote from Sherlock Holmes: "When you have eliminated the impossible, whatever remains, however improbable, must be the truth" - and that's differential diagnosis in a nutshell: list all the possibilities, then systematically eliminate them until there's only one left.

What makes the show so interesting, though, is that Dr. House and his team don't see patients with one symptom. The patients who get referred to them often have multiple symptoms; people who suddenly have a whole bunch of different and apparently unrelated things going wrong, all at the same time:



Video clip: A scene from “House MD”, in which House and the team work through a diagnostic plan for a patient presenting with seven unrelated symptoms.



The specific detail that I love about this clip is the probability. The patient has seven separate symptoms that, together, could only be explained by an incredibly rare cardiac infection - so rare it's a one in ten million shot. Or, the patient has two separate conditions which, between them, explain all of the symptoms... much more common conditions, a thousand to one, so House figures the likelihood of suddenly developing two of them at the same time - what doctors refer to as co-morbidity - is only one in a million - which is still ten times more likely than Dr. Foreman's cardiac infection theory.



And, of course, there are the scenarios where one problem is the underlying cause, and the other problems are the symptoms: if you identify, and fix, the cause, then all the symptoms disappear. One really common example of this happens to a lot of people as they get older; a worried family brings them to the doctor: they're depressed, they're struggling to remember things, they're withdrawn, uncommunicative, problems with their balance - maybe they've even started stumbling and falling over more often. Oh, and they've also started turning the TV up really loud so maybe they're having problems with their ears... well, turns out that when you start to lose your hearing, it has all kinds of side-effects. You stop socialising 'cos you can't follow a conversation. You can't remember things because you never heard them in the first place, you get really depressed 'cos you're feeling lonely and cut off - and it turns out that proprioception, the sense humans use to help us balance and know where we are in relation to our environment? That's weirdly linked to our sense of hearing - almost like a subconscious sonar system.



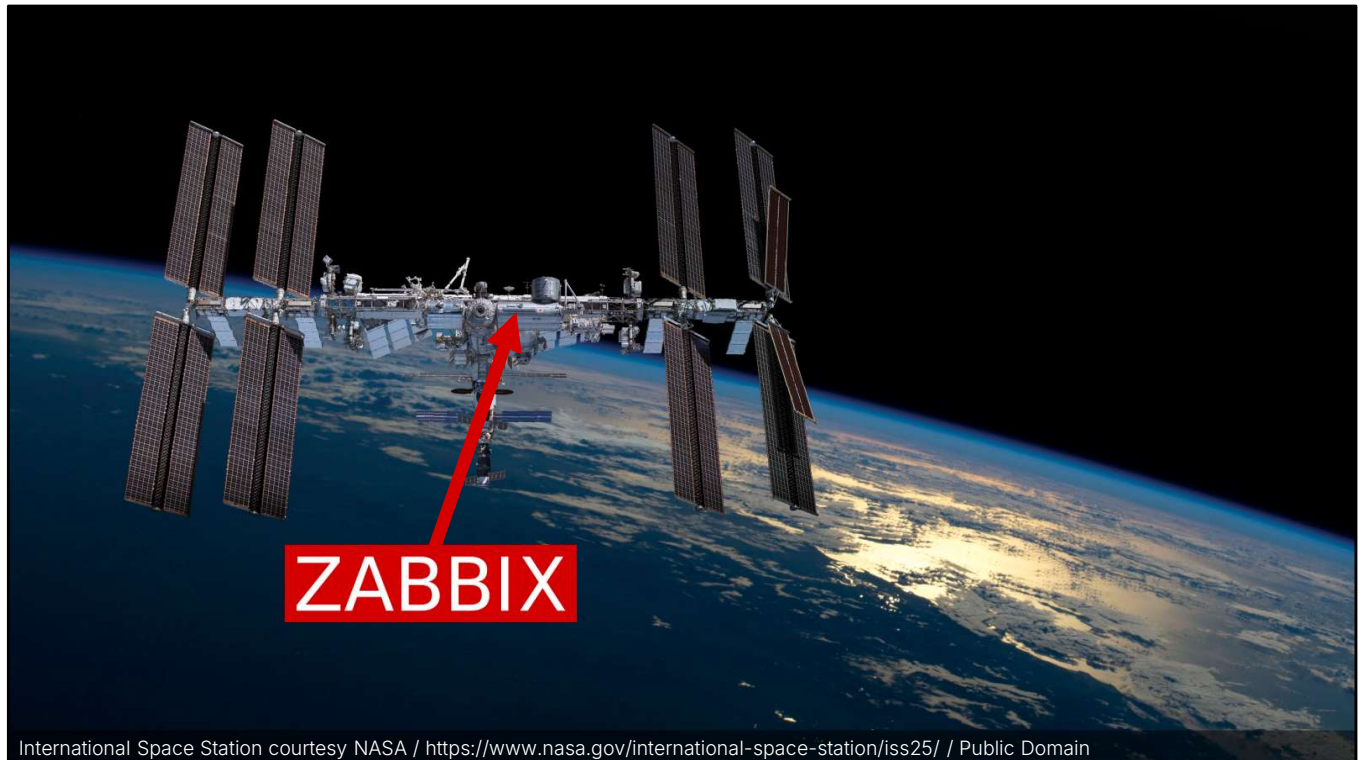
Fix the cause - usually with hearing aids - and the rest of the symptoms disappear. Kinda like how if there's a power outage in your data centre, that might not be the first thing you notice. The first sign is probably that a website's not responding, or somebody can't open the mobile app. Somebody else says their reports are timing out, here's a ticket saying the intranet is down... oh, and there's the alert saying none of the servers are responding to pings... and you know what Zabbix calls those kinds of problems? Symptoms and causes. Yep, straight out of the medical textbook.

Turns out there are a lot of parallels between medical diagnosis and systems monitoring... up to a point. And that point is the moment where House says to the team:

**"Tests take time...
treatment's
quicker"**

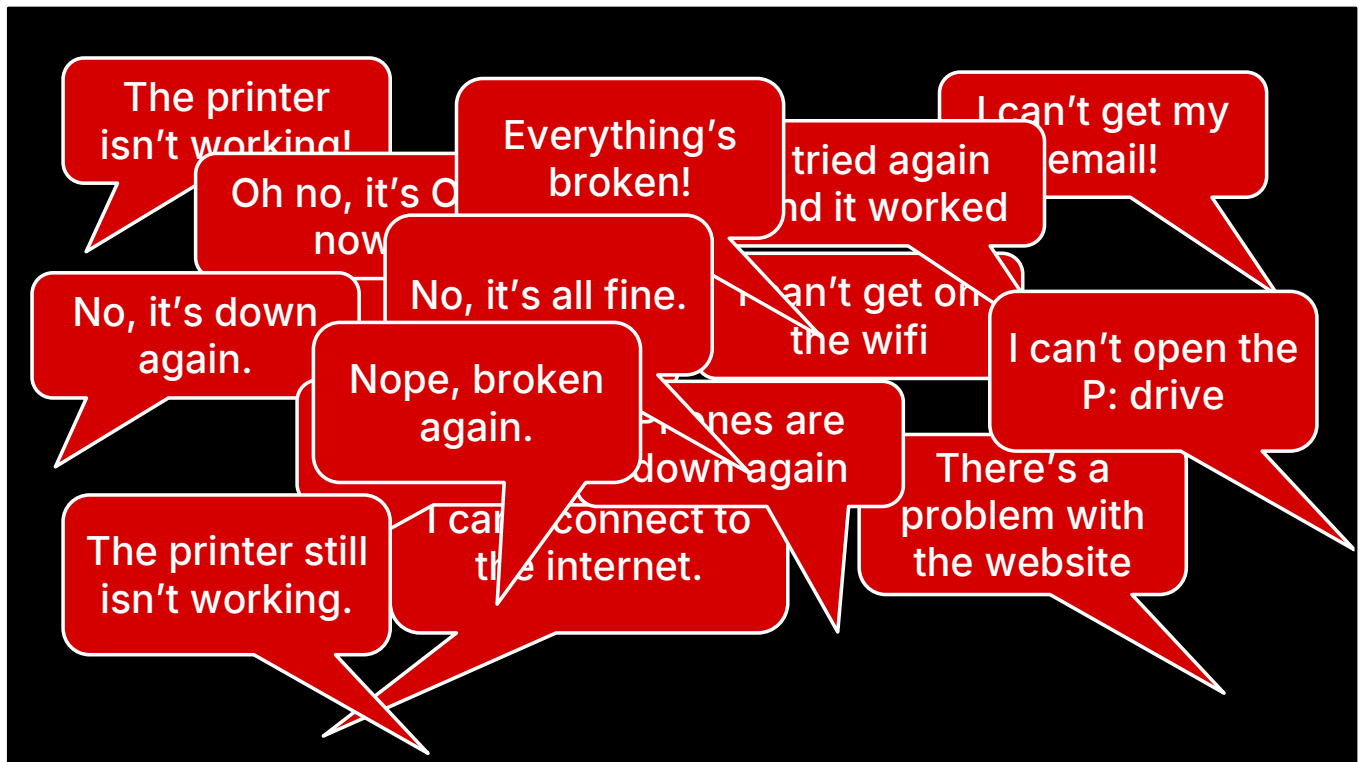
"tests take time; treatment's quicker". One of the recurring dramatic themes in the show is the moments where the team has a theory, and a treatment plan: if their theory is correct, the patient will recover - but if their theory is wrong, the treatment they're proposing could make the patient sicker - or even kill them. And in medicine, if you accidentally kill the patient, you can't just restore them from a backup and bring them back online - so on the one hand, you've got to be much more careful about how you validate your hypotheses. On the other hand, if you do nothing the patient is probably going to die, so you don't have the time to run every available test.

I know there's folks out there using Zabbix to monitor some pretty critical infrastructure: air traffic control, transport networks... even in space;



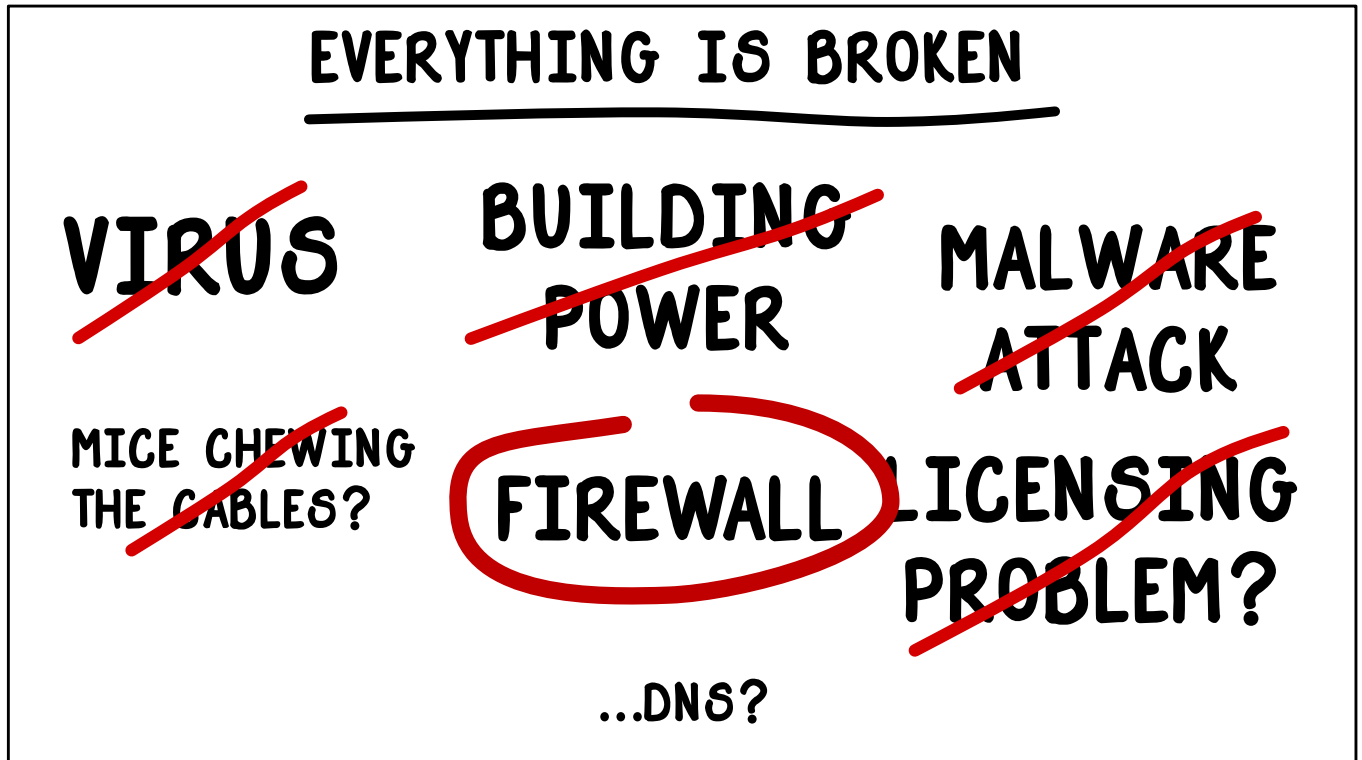
International Space Station courtesy NASA / <https://www.nasa.gov/international-space-station/iss25/> / Public Domain

there's a Zabbix proxy running on the Columbus module, the European Space Agency's research laboratory that's part of the International Space Station. Personally, I've never worked on anything quite that critical - although I've certainly worked for a few clients who thought that their website being down was a matter of life and death. But I've definitely been on call for a few incidents that felt like an episode of House; that same pattern, one or two minor symptoms that quickly escalate into something really quite serious.



One of them happened about twenty years ago, when I was working as "Head of IT" for a company in London - and being a pretty small company, Head of IT meant I did absolutely everything - web programming, database, network, firewall, managing the Microsoft Exchange server...

One morning, around 11am, I get a call: somebody's computer can't connect to the internet. Big deal. That happens all the time. Then another one. Then somebody else can't print. Somebody else can't log in. Oh, no, it's OK, it's working again now... no, it's gone again... Some folks can't get to their email. Not everybody, just some of them. Over the next few hours, just about every device in the entire company starts showing weird behaviour. No error messages, nothing in the log files... they're just being weird.



So, differential diagnosis. Maybe - just like in a medical show - it's a virus? Problem with the building power supply? Malware? Firewall? Maybe it's a licensing problem? Maybe there's... I don't know, mice chewing the cables? We're getting desperate here. Or maybe it's DNS.

Probably not a virus – we've got PCs, Macs and printers all affected. Building power? Everything else is fine. Mice? Nobody's seen any mice. Probably not mice. Licensing? Nah, we're fully paid up. Malware attack? No signs of any suspicious activity. Firewall? That's definitely connected to everything that's going wrong... so how do we rule out the firewall? Let's replace it. So we order a new firewall, replicate the configuration by hand – just in case the config is corrupted – plug it in... nope, it's not the firewall. But hang on...

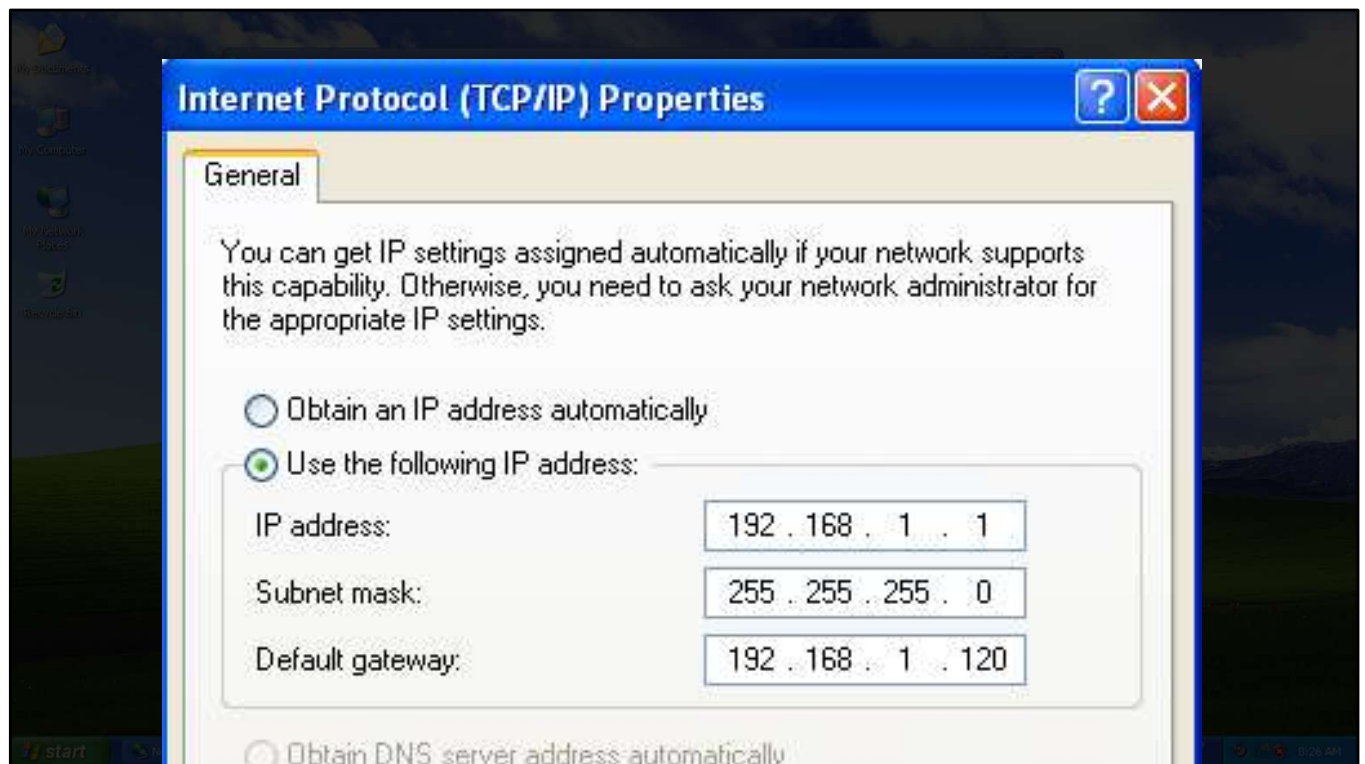
```
Pinging 192.168.1.1 with 32 bytes of data:  
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64  
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64  
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64  
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64  
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64  
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Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
```

So we unplug it, and just as we're about to plug the old one back in, my buddy Scott notices something really weird. The firewall is configured as the default gateway on the network - 192.168.1.1. We've got a continuous ping running on one of the servers - and when we unplugged the firewall, the ping didn't stop. It's still responding.

Yeah, you got it. The firewall's not the only thing that's responding to the pings.

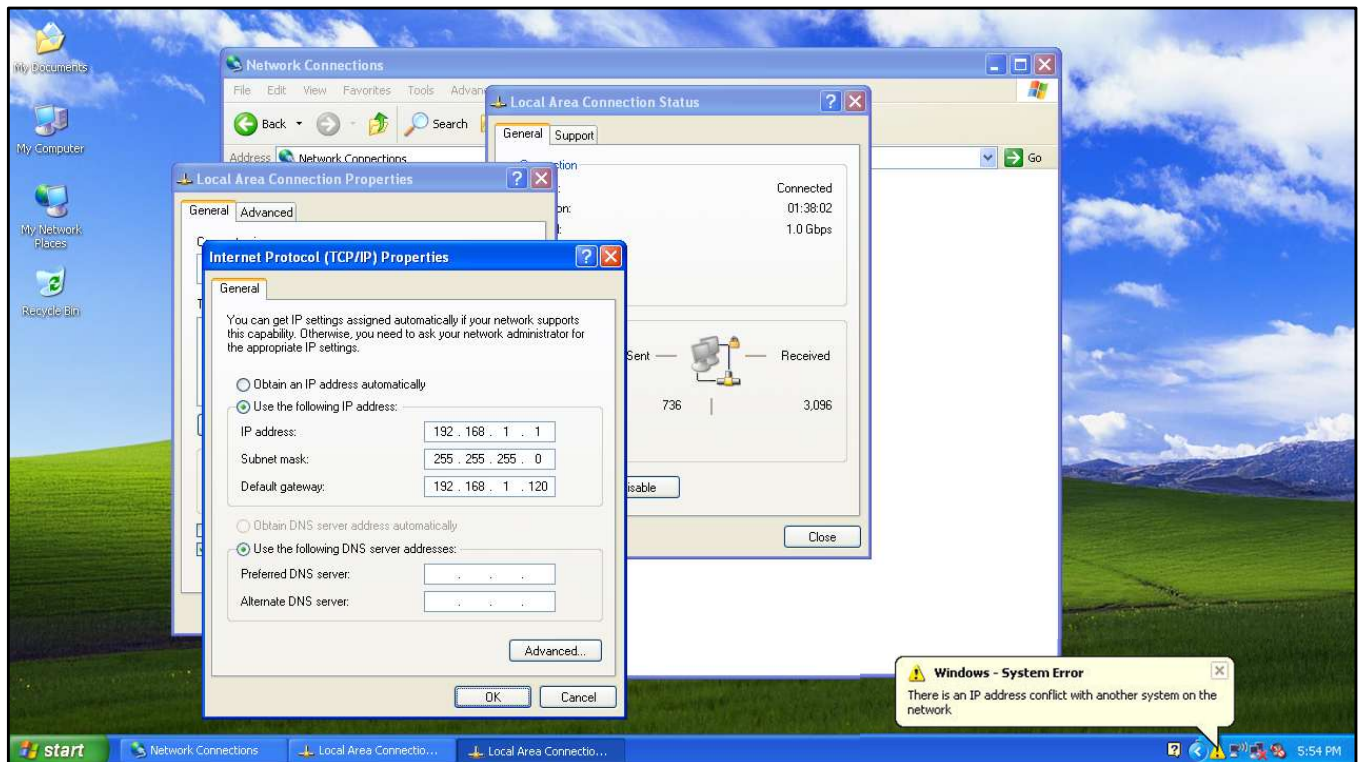
Earlier that day, one of the camera operators in our video studios - let's call him Nick - had had a problem with their workstation, and knew just enough about TCP/IP networking that they tried to fix it themselves. Which, of course, regular users shouldn't have permission to do: that's a terrible idea.

Except... the combination of BlackMagic camera control software and Adobe OnLocation that we were using in the studios only worked reliably if the user had local administrator permissions. Sure, we'd complained, we had an open support ticket... but in the meantime, we had a business to run, and if the only way to run the recording sessions was to give the studio team local admin permissions, that's what we had to do.



Nick had unwittingly got the IP address and default gateway the wrong way around. So now there were two devices on the network with the same IP address.

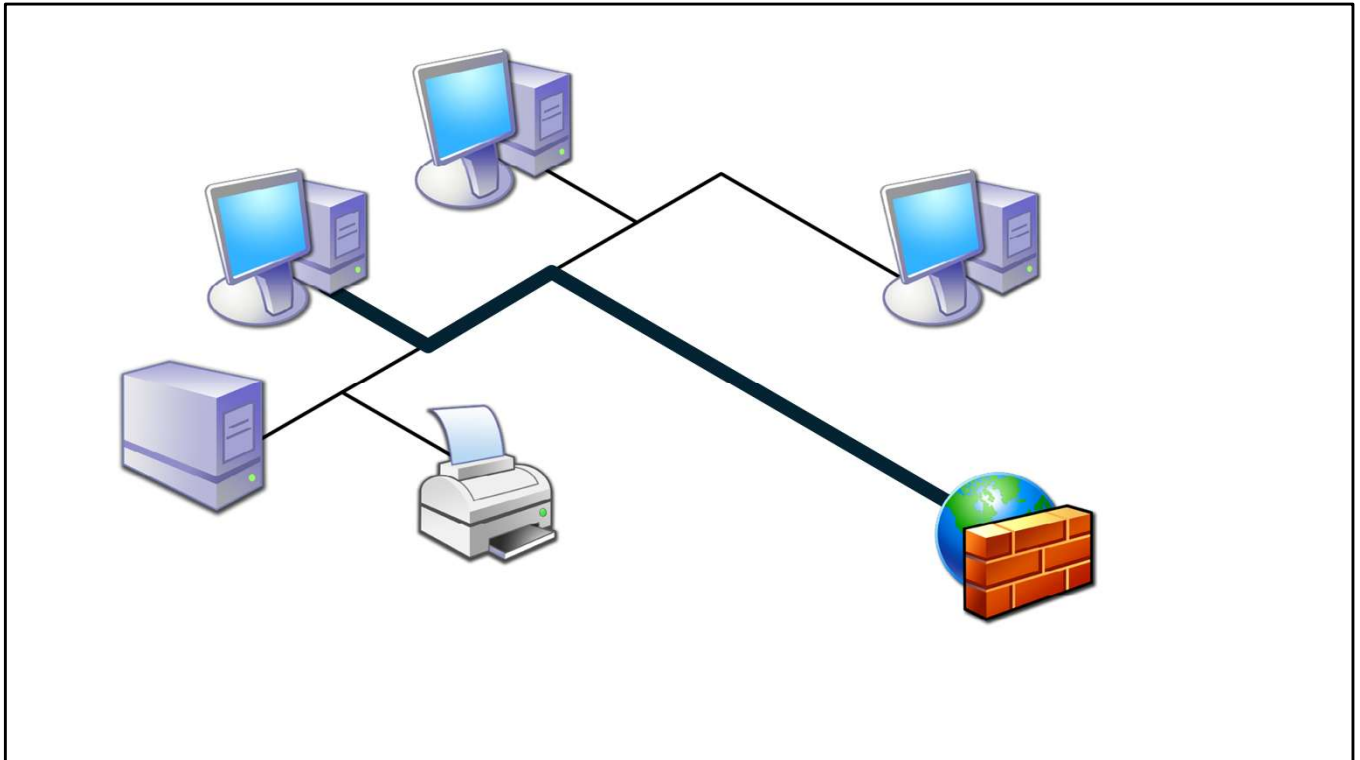
Dylan Beattie: The Curious Incident of the Log in the Downtime.



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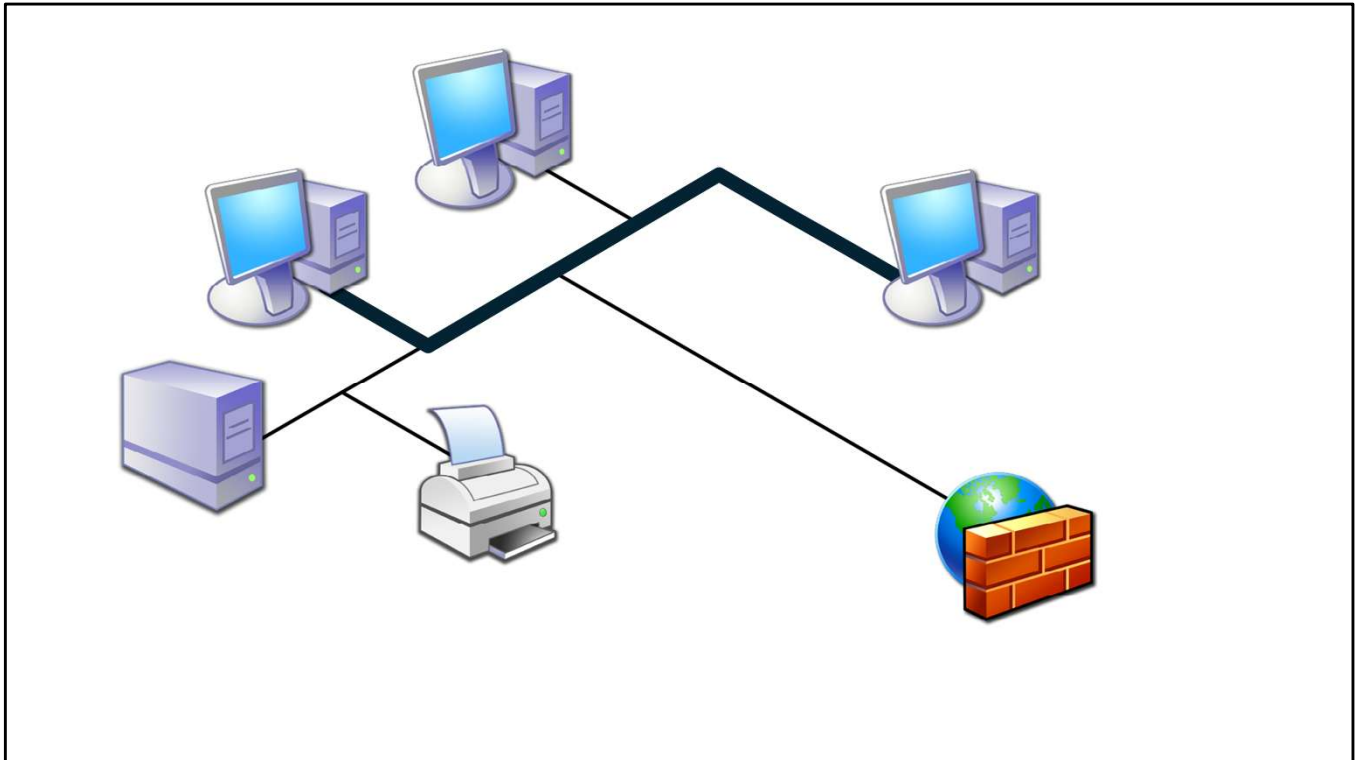
Sure, Windows kept popping up a warning on that machine about an IP address conflict... but Nick didn't think it was important, and didn't mention it to anybody. Besides, we were clearly **REALLY BUSY** dealing with the **MASSIVE PRODUCTION OUTAGE** and didn't have time to worry about one little network problem in the video studio...

Anyway, the outcome of all of this is that every few minutes, every device on the network was saying "hey, I need to send these packets to 192.168.0.1 - which machine is that?" This is called address resolution; the protocol that makes it work is called the Address Resolution Protocol, normally shortened to ARP, and inside your laptop, and your phone, and just about every piece of networked hardware on this planet, there's a thing called an ARP cache. See, your computer doesn't look the address up every time. When it gets an answer, it'll remember it - on Windows, the ARP cache is between 15 and 45 seconds.



What does that mean in practice? It means that when your computer tries to talk to the internet, it's going to say "hey, where shall I send these packets?" and, because there are two different machines on the network, both claiming that THEY'RE the default gateway, you'll get one at random. If you get the firewall? Awesome. Your computer works. If you get Nick's workstation? Sorry, bub. No internet for you... but then if you don't DO anything for that random interval - 15 to 45 seconds - that record will get marked as Stale, and next time you try to talk to the internet, it'll ask again: "hey, where do I send these packets?" - and you got a 50/50 chance of getting the right answer. And if you do... hey, everything works! Until you answer the phone, or go to get a cup of coffee, and the cache times out, and it all starts again.

Now, unless there's anybody out there who was working with me in London in about 2008 - anybody? - none of you folks will have noticed that particular incident.



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NOVEMBER						2025
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But I'm sure a few of you remember what happened on 18th November 2025. You might not remember the exact date... first thing I heard about it was one of my clients pinging me to say that their website was down. Which was weird, because I had two different sets of monitoring pointing at that website - it was hosted on Azure, I had Azure's AppInsights running, AND a third-party endpoint monitoring service. In theory, anything goes down, I find out about before the client does - that's the whole point of monitoring, right?

So I log in to their dashboard... this is interesting. They're also having a bad day. So, probability time: either our site's had an outage at the EXACT SAME TIME that our monitoring platform ALSO had an outage... or there's a common underlying cause.

But then I take a closer look at the actual error message we're getting... and notice something interesting. Do any of you folks out there use Cloudflare in front of your websites? Yeah, you do. About half the internet uses Cloudflare.

I've seen THIS error a bunch of times

Error 521

Web server is down

You

Browser

Working

CloudFlare

Working

www.my-important-site.com

Host

Error

What happened?

The web server is not returning a connection. As a result, the web page is not displaying.

What can I do?

If you are a visitor of this website:

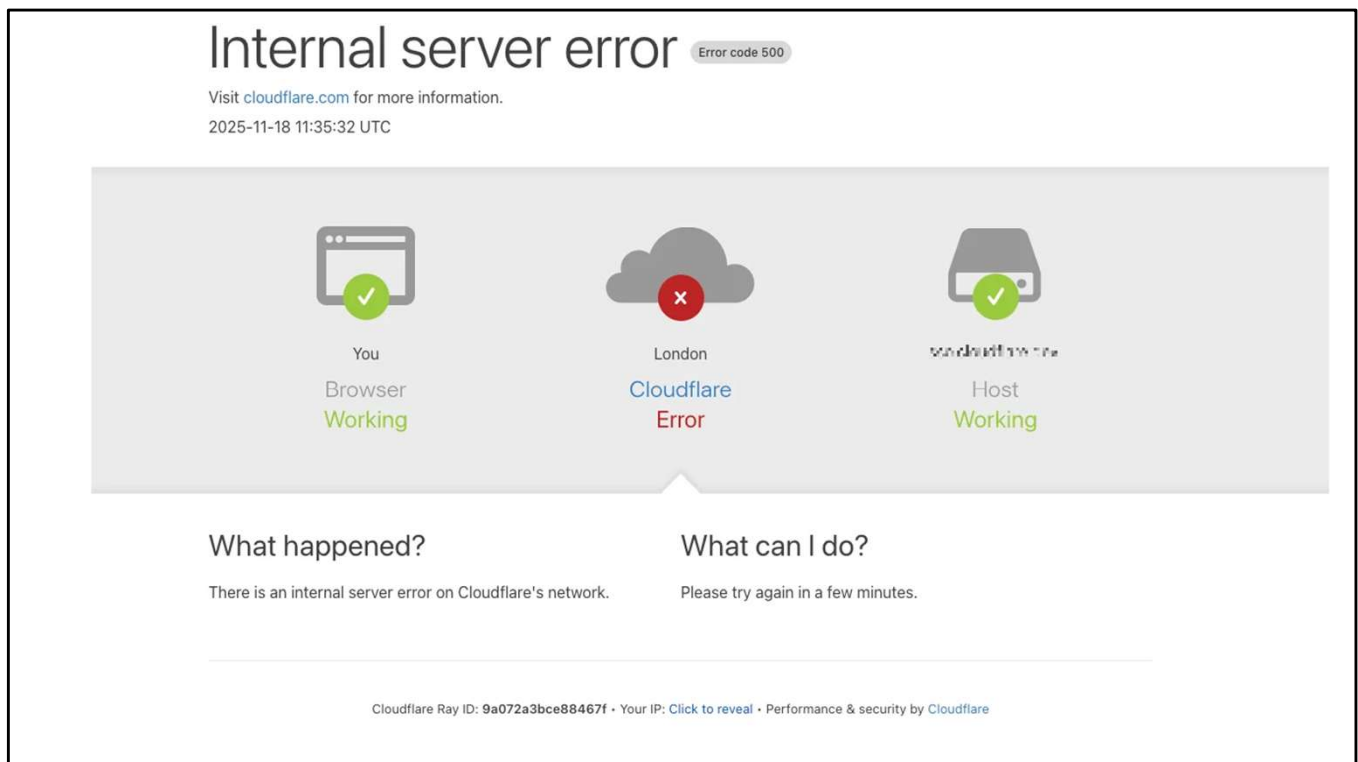
Please try again in a few minutes.

If you are the owner of this website:

Contact your hosting provider letting them know your web server is not responding. [Additional troubleshooting information.](#)

- but THIS one?

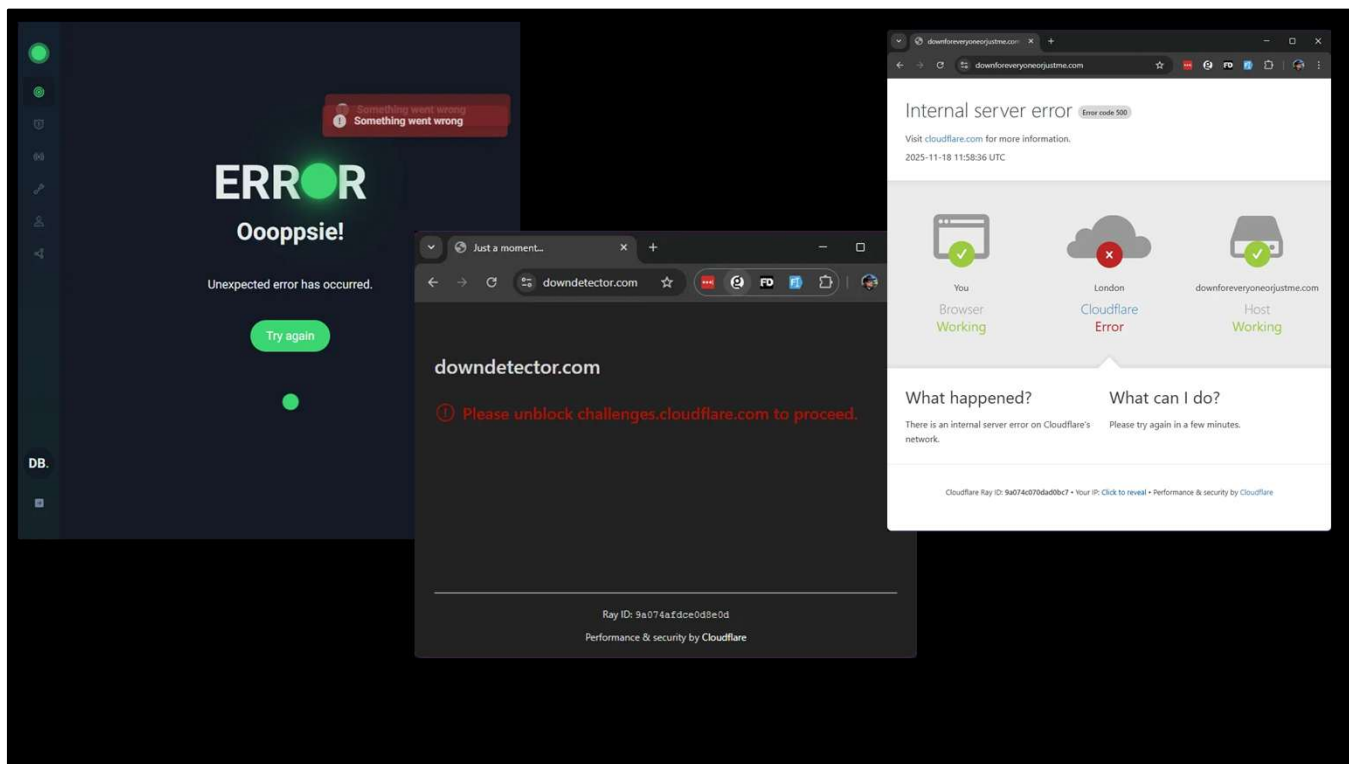
Dylan Beattie: The Curious Incident of the Log in the Downtime.



I haven't seen that one before.

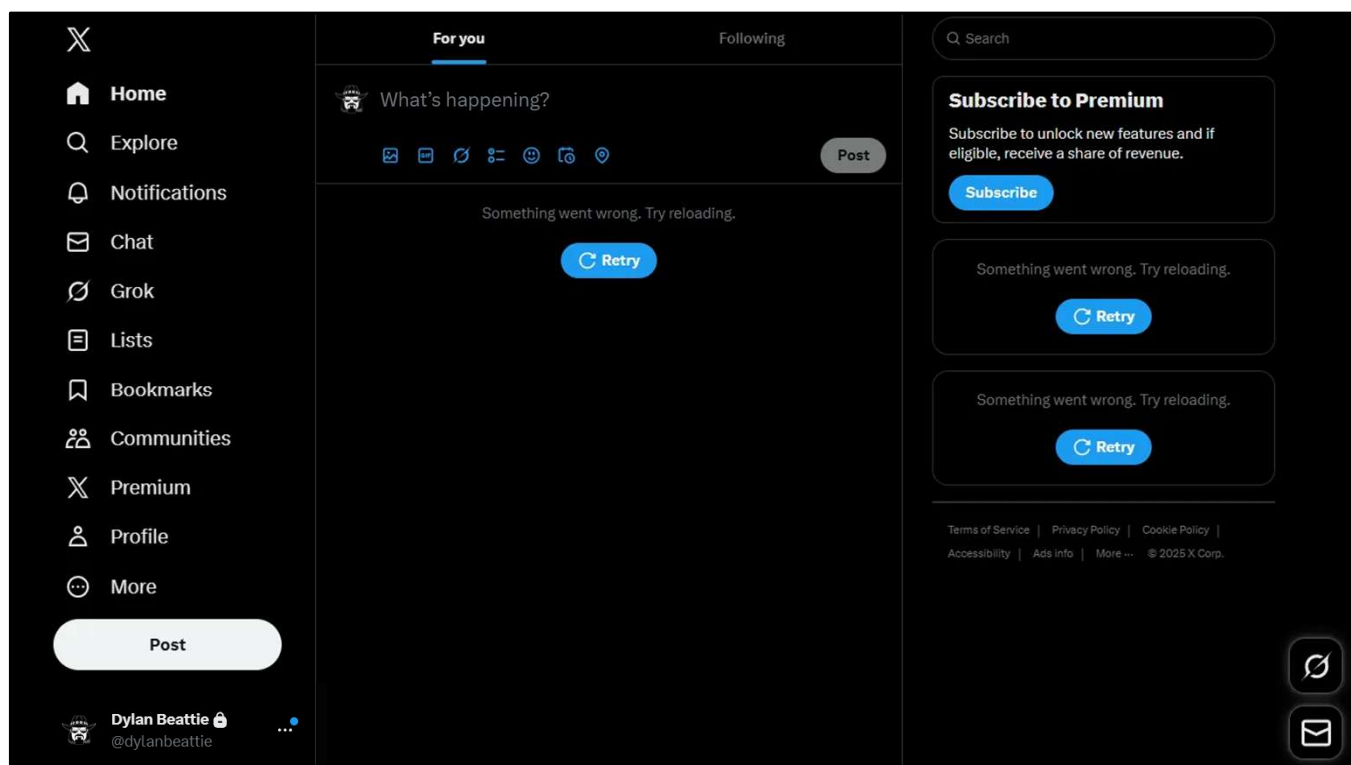
Yep... Cloudflare's the problem. And when I said about half the internet uses Cloudflare, I wasn't kidding... because when Cloudflare has a bad day, it takes out, well, about half the internet.

Dylan Beattie: The Curious Incident of the Log in the Downtime.



My clients' sites were down. UptimeRobot was down. DownDetector.com was down. DownForEveryoneOrJustMe.com was down. I even logged in to Twitter to see if anybody was reporting it... yep, Twitter was down.

Dylan Beattie: The Curious Incident of the Log in the Downtime.



When I say "down", of course, I don't mean completely offline - these are all distributed applications with multiple backend APIs, caching, proxies, all kinds of things, so when those backend APIs become unreachable you get a really interesting insight into how those sites actually work.

Oh, and apparently Zabbix Cloud was absolutely fine, so I'm guessing there's a fair few folks here today who remember the Cloudflare incident because they got notified about it just like any other outage, quickly figured out it was nothing to do with them, and spent the rest of the morning drinking coffee and waiting for Cloudflare to come back up.

Dylan Beattie: The Curious Incident of the Log in the Downtime.



There was a similar incident back in 2021 - a company called Fastly deployed a broken Varnish configuration that took down dozens of websites - BBC News, Amazon, Reddit, GitHub, Stack Overflow - and I'm sure you folks remember the

Dylan Beattie: The Curious Incident of the Log in the Downtime.



Photo © Reuters/Bing Guan via theguardian.com

CrowdStrike outage in July 2024 that took out nearly ten million Windows PCs. Governments, airports, banks, hospitals were all affected; the total cost of that outage was estimated to be around ten billion dollars.

The irony, of course, is that those services - Cloudflare, Fastly, CrowdStrike - the only reason anybody was using them in the first place was to improve reliability. I remember describing the CrowdStrike incident to somebody at the time: imagine there's a company that sell fire safety equipment



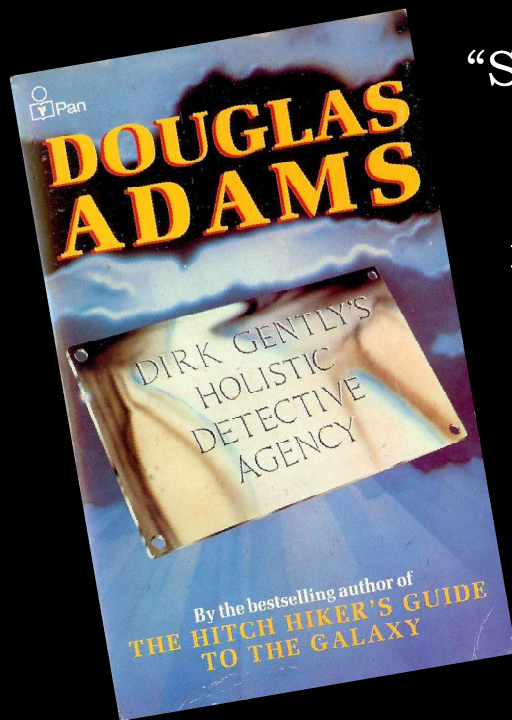
https://www.reddit.com/r/Wellthatsucks/comments/dbake8/fire_extinguisher_caught_fire/

- sprinklers, smoke detectors, halon systems. They've got equipment installed in offices all over the world, millions of companies are relying on them - and then at 4am one day all of their smoke alarms suddenly catch fire and thousands of buildings burn down.

I'm sure that on that fateful day, more than one person must have said "But CrowdStrike can't be causing the problem... that's impossible!" - and that's the real challenge. Differential diagnosis only works if you can eliminate the impossible... but this is tech. We do six impossible things every day before breakfast. We send electrons flying around the planet at the speed of light, carrying emails, pictures, video, music; we have algorithms that can index all the world's information, we have language models that can write Python code and translate jokes from Urdu into Swahili, every single one of you right now has a high-definition wireless supercomputer in your pocket that's become such a part of everyday life that we don't even realise it's magic any more, we just use it to share cat pictures and watch TikToks.

And things that we thought were impossible? They happen all the time. The firewall we just unplugged is still responding to pings. The server's been shut down but the website's still up. It can't be DNS. No way. We checked. That's impossible... but yep, turns out it was DNS.

And so I want to give the last word today to my favourite fictional detective.



“Sherlock Holmes observed that once you have eliminated the impossible then whatever remains, however improbable, must be the answer...

I, however, do not like to eliminate the impossible.”

Created by the late, great Douglas Adams, Dirk Gently was the world's only holistic investigator, and as he once memorably said:

“Sherlock Holmes observed that once you have eliminated the impossible then whatever remains, however improbable, must be the answer. I, however, do not like to eliminate the impossible.”



(SCAN HERE FOR SOCIALS
AND SLIDES AND WHATNOT)



Liels paldies!
(thank you very much!)

Dylan Beattie
dylanbeattie.net/zabbix

ZABBIX
SUMMIT 26
OCTOBER 7 • 9, 2026
RIGA • LATVIA

Zabbix, thank you very much. I'm Dylan Beattie, you can find me on the internet at dylan beattie dot net; I used to use this final slide to share social media links but social media's all horrible now so I have a mailing list - I know, old school, eh? - so if you want to hear about things I'm working on and places I'm going and courses I'm running, scan the QR code and sign up. And if you don't? Well, that's OK too. You folks take it easy, enjoy the rest of your day, and I'll see you around.